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(12) **United States Patent**
Sollie et al.

(10) **Patent No.: US 12,415,331 B2**
(45) **Date of Patent: *Sep. 16, 2025**

(54) **SHEET WITH CUSHIONING INSERTS**

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(73) Assignee: **Pratt Corrugated Holdings, Inc.**, Brookhaven, GA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **17/177,908**

(22) Filed: **Feb. 17, 2021**

(65) **Prior Publication Data**

US 2022/0032572 A1 Feb. 3, 2022

Related U.S. Application Data

(60) Provisional application No. 63/059,001, filed on Jul. 30, 2020.

(51) **Int. Cl.**

B65D 81/03	(2006.01)
B32B 3/08	(2006.01)

(Continued)

(52) **U.S. Cl.**

CPC ***B32B 3/08*** (2013.01); ***B32B 3/30*** (2013.01); ***B32B 7/12*** (2013.01); ***B32B 29/005*** (2013.01);

(Continued)

(58) **Field of Classification Search**

CPC B65D 81/03; B65D 81/051; B65D 81/02; B65D 81/022; B65D 81/025; B65D 81/027; B32B 3/08; B32B 3/30

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Primary Examiner — Nathan J Newhouse

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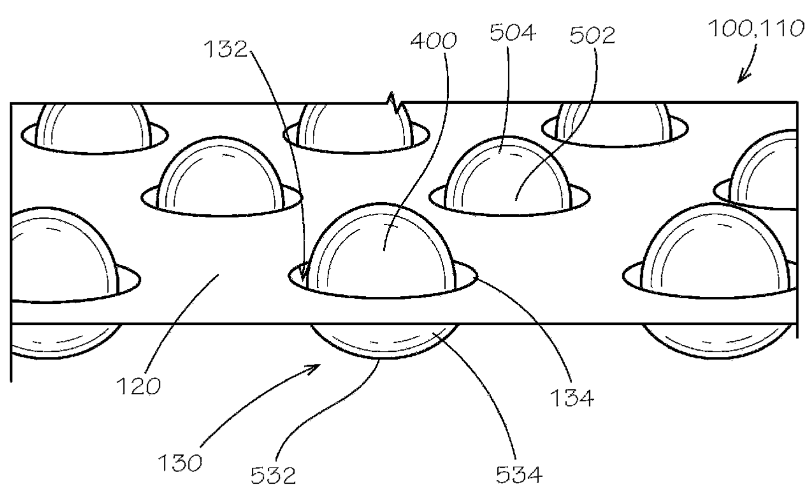
(57) **ABSTRACT**

Example aspects of a cushioned sheet, a cushioned mailer, and a method of forming a cushioned sheet are disclosed. The cushioned sheet can comprise a first layer comprising a first base and a plurality of first sockets extending from the first base, each of the first sockets spaced apart from adjacent ones of the first sockets; a second layer coupled to the first layer by an adhesive, the second layer comprising a second base and a plurality of second sockets extending from the second base, each of the second sockets aligned with a corresponding one of the first sockets to define a void therebetween; and a plurality of cushioning inserts, each of the plurality cushioning inserts substantially spherical in shape and received in a corresponding one of the voids.

18 Claims, 12 Drawing Sheets



(51) **Int. Cl.**

B32B 3/30	(2006.01)
B32B 7/12	(2006.01)
B32B 29/00	(2006.01)
B32B 37/00	(2006.01)
B32B 37/10	(2006.01)
B65D 27/14	(2006.01)
B65D 81/02	(2006.01)

(52) **U.S. Cl.**
CPC ***B32B 37/0053*** (2013.01); ***B32B 37/10*** (2013.01); ***B65D 27/14*** (2013.01); ***B65D 81/02*** (2013.01); ***B65D 81/03*** (2013.01); *B32B 2307/558* (2013.01); *B32B 2439/40* (2013.01)

(58) **Field of Classification Search**
USPC 229/80
See application file for complete search history.

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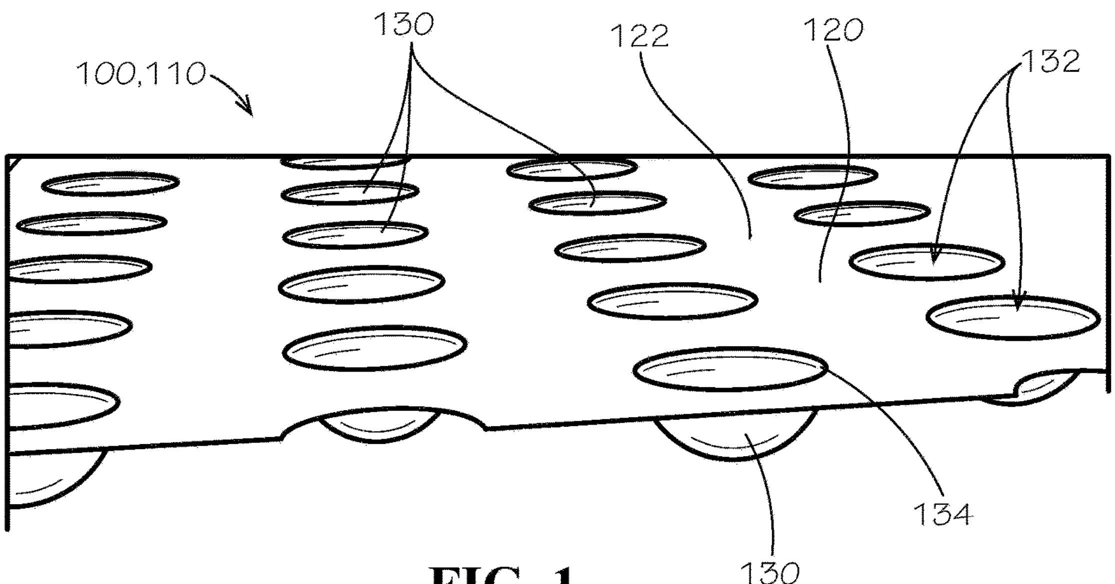
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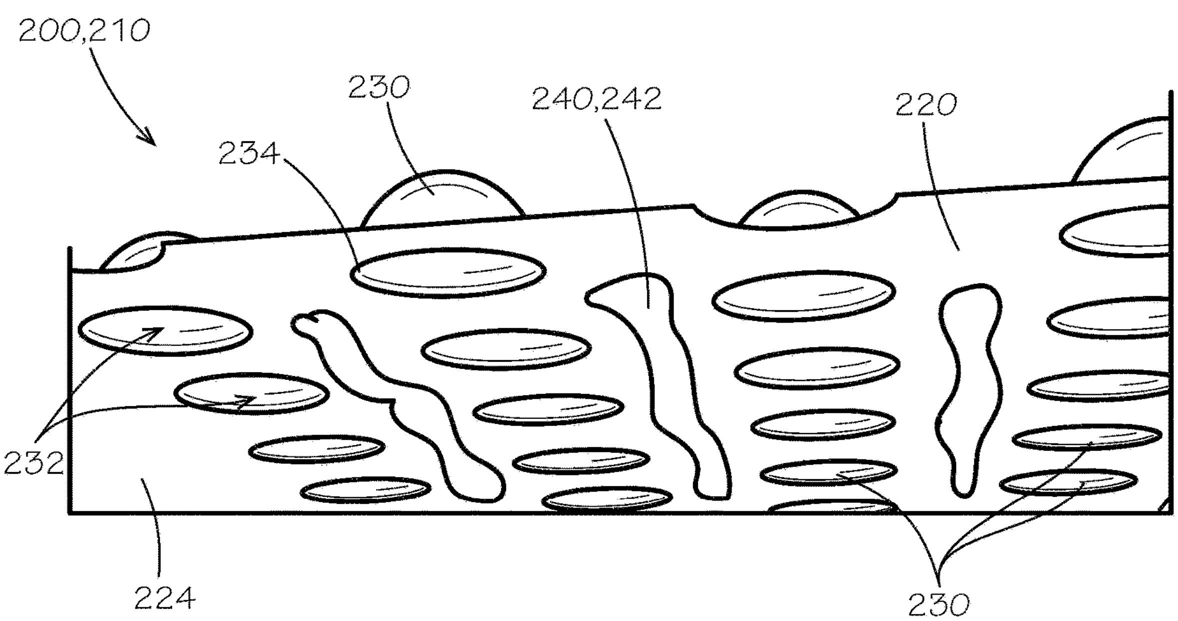
FIG. 1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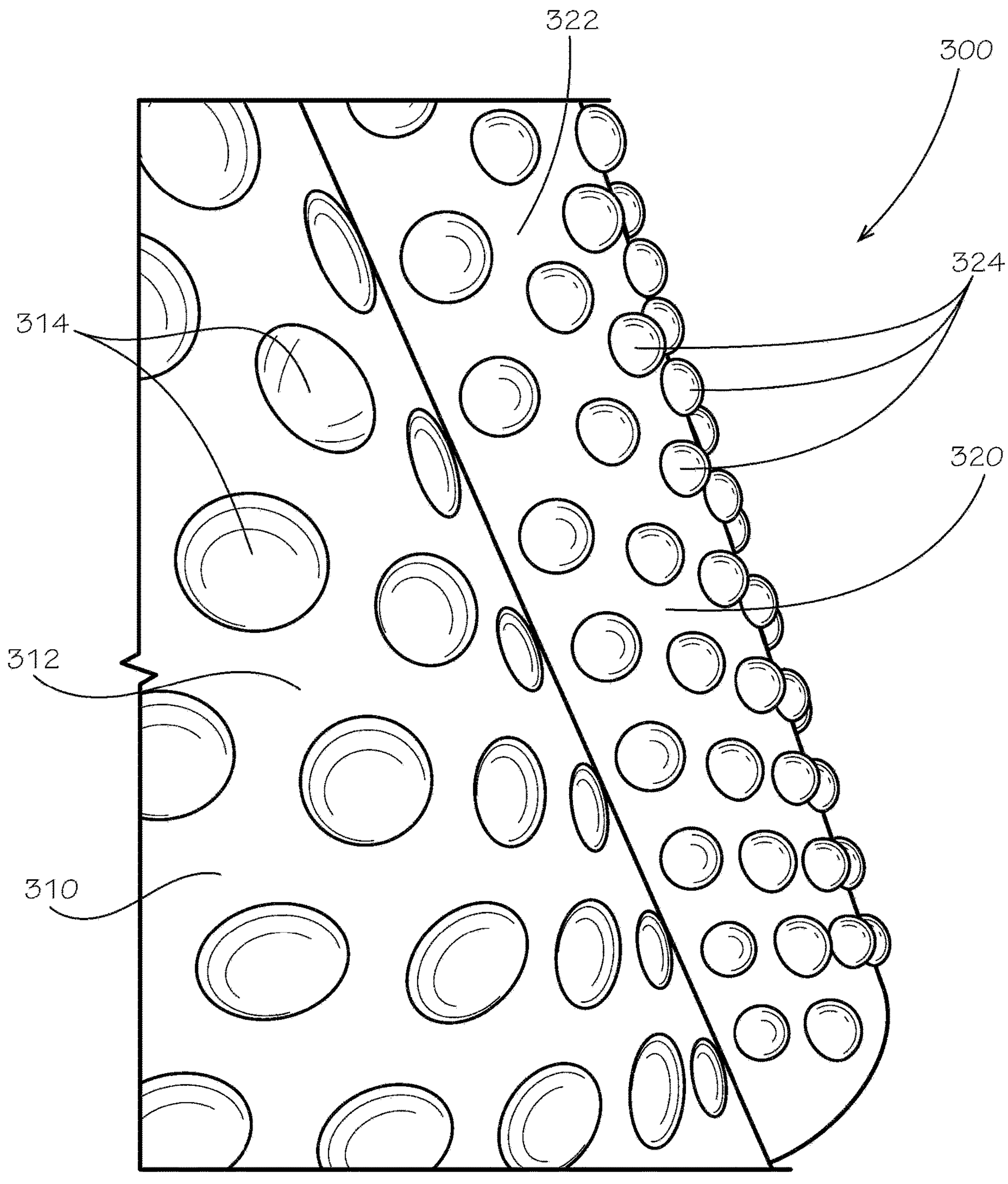

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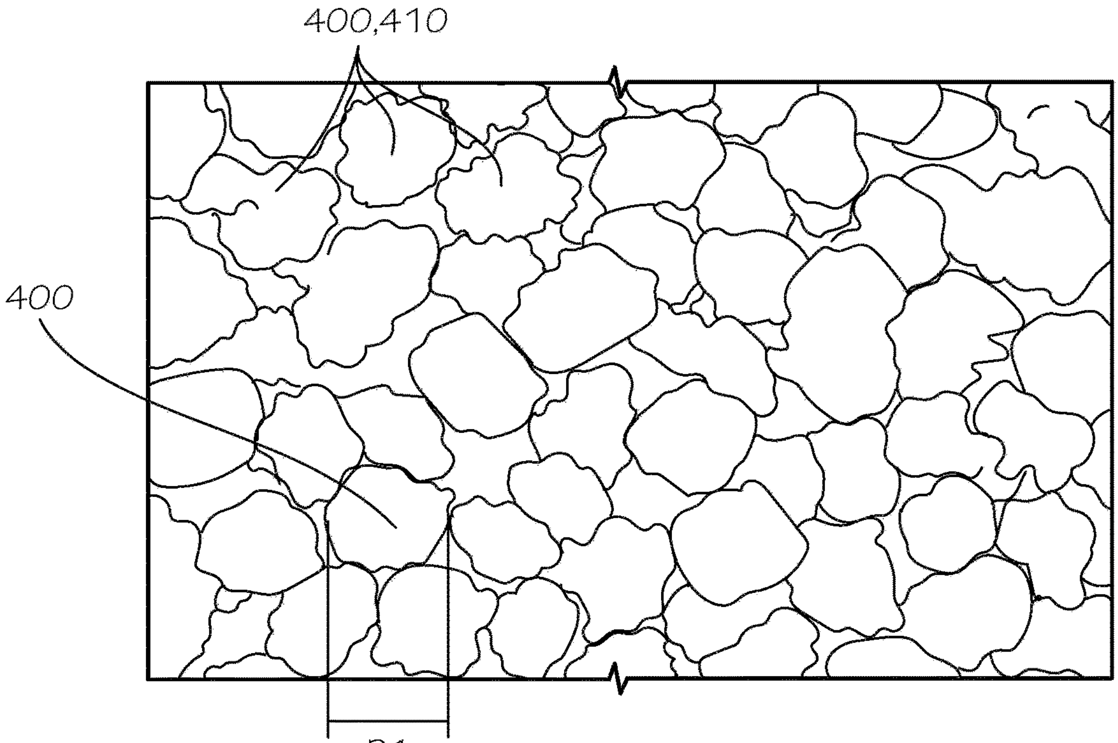

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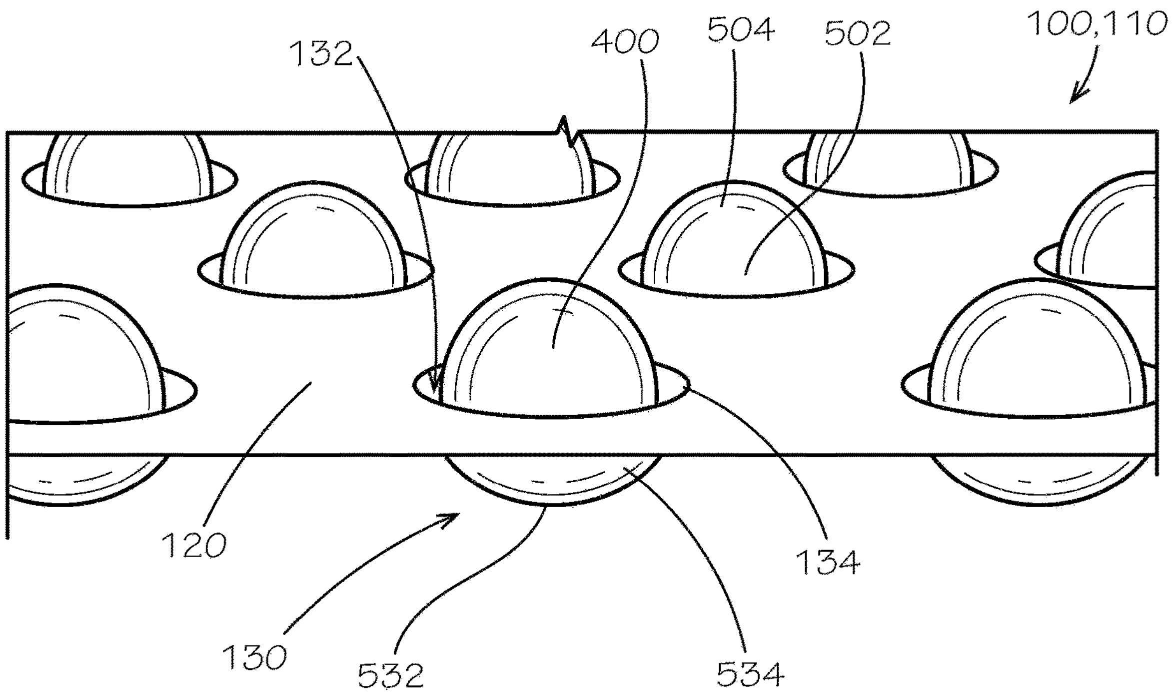

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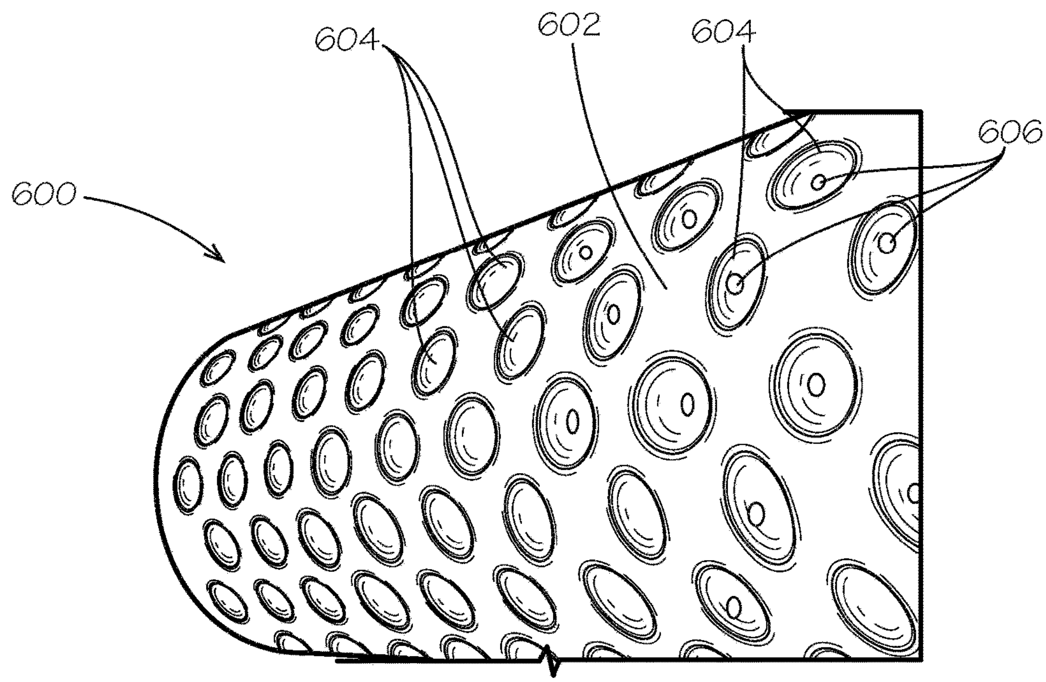

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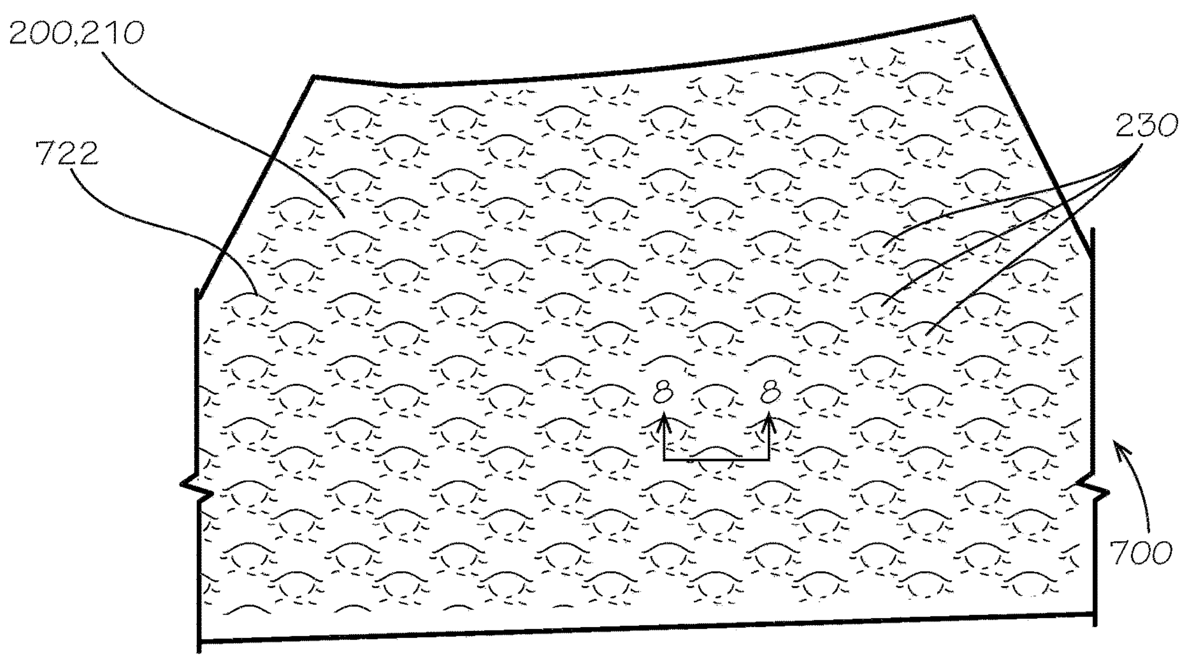

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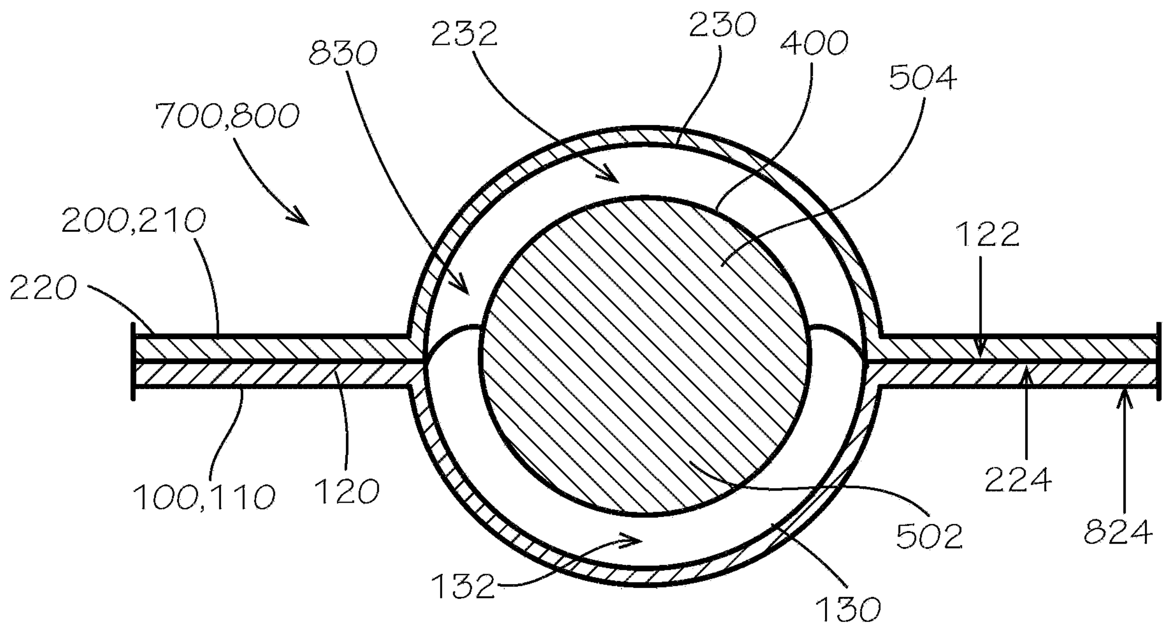

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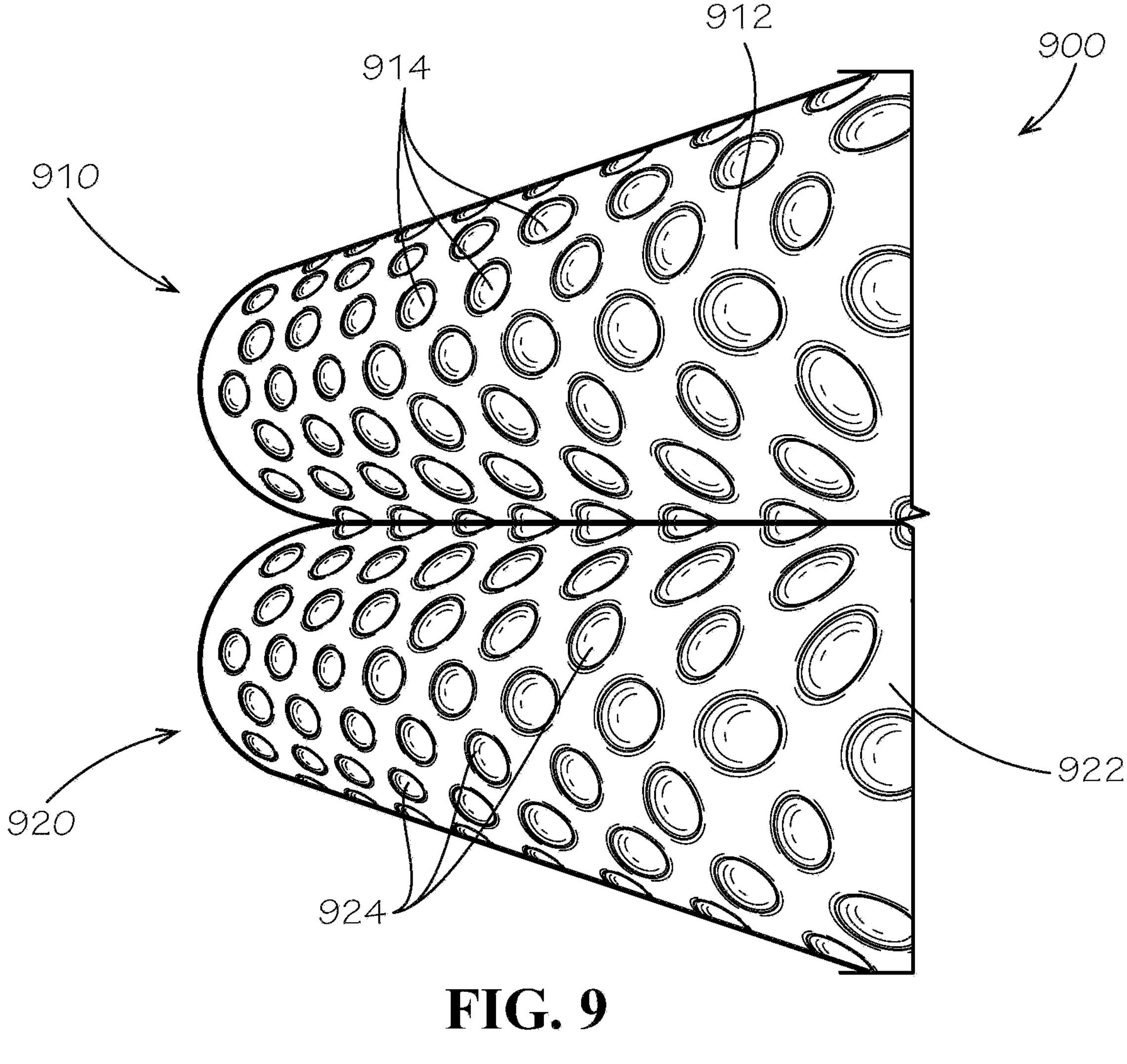

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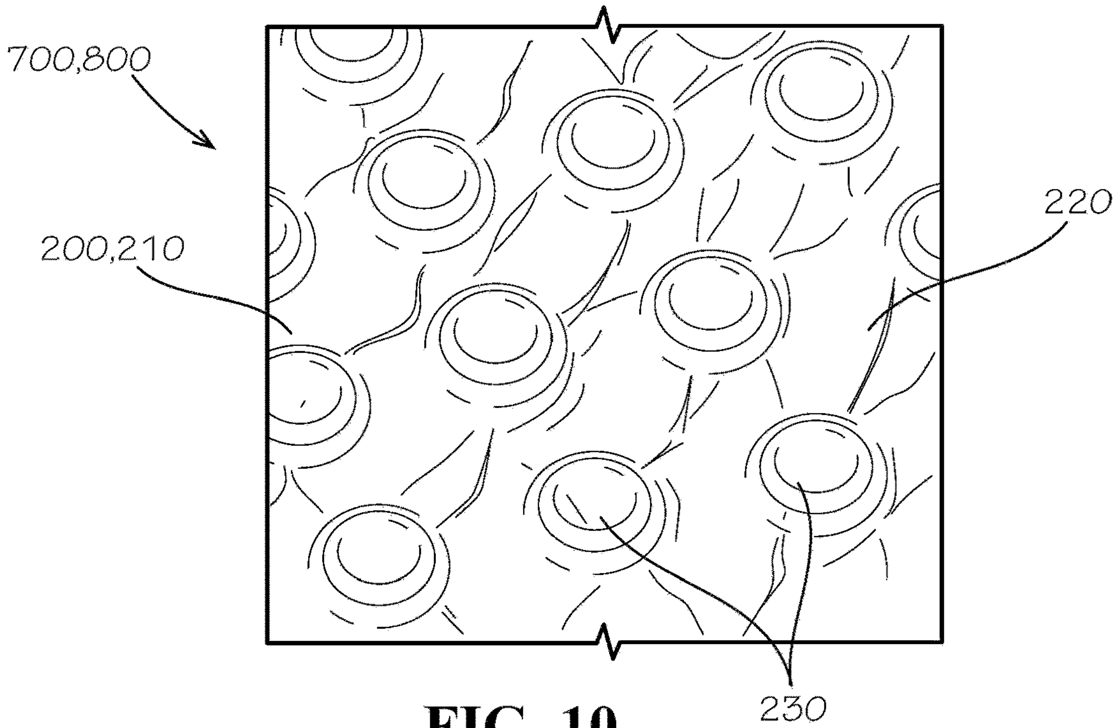
FIG. 9



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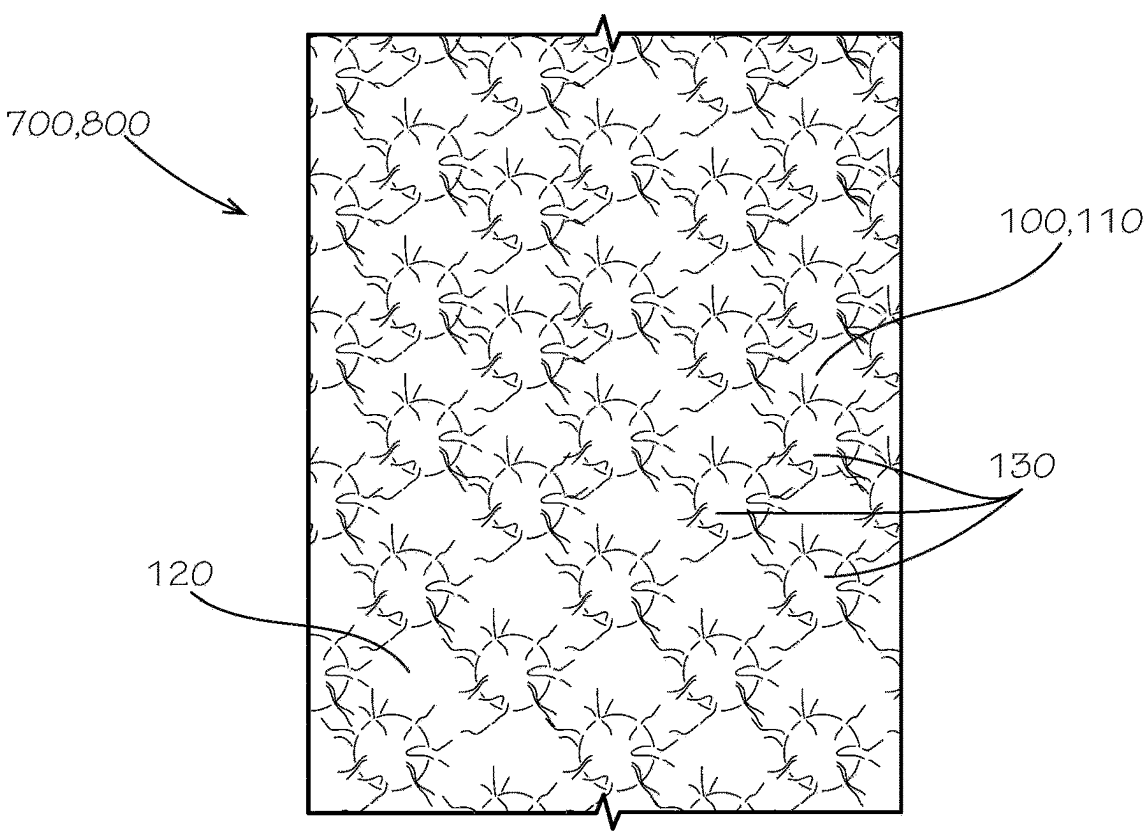


FIG. 11

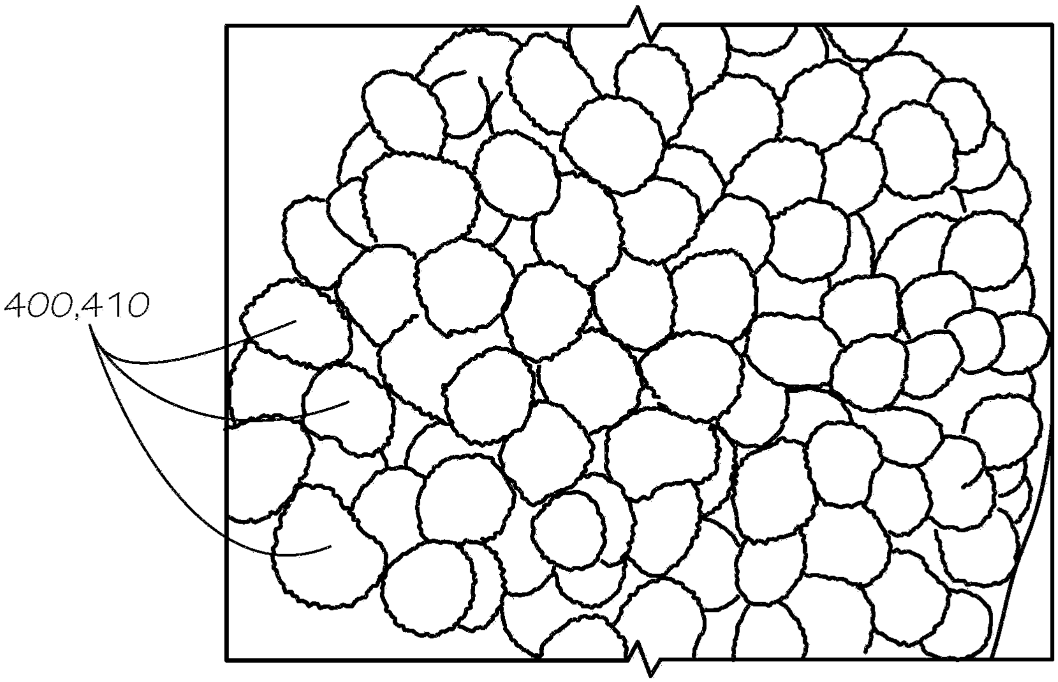
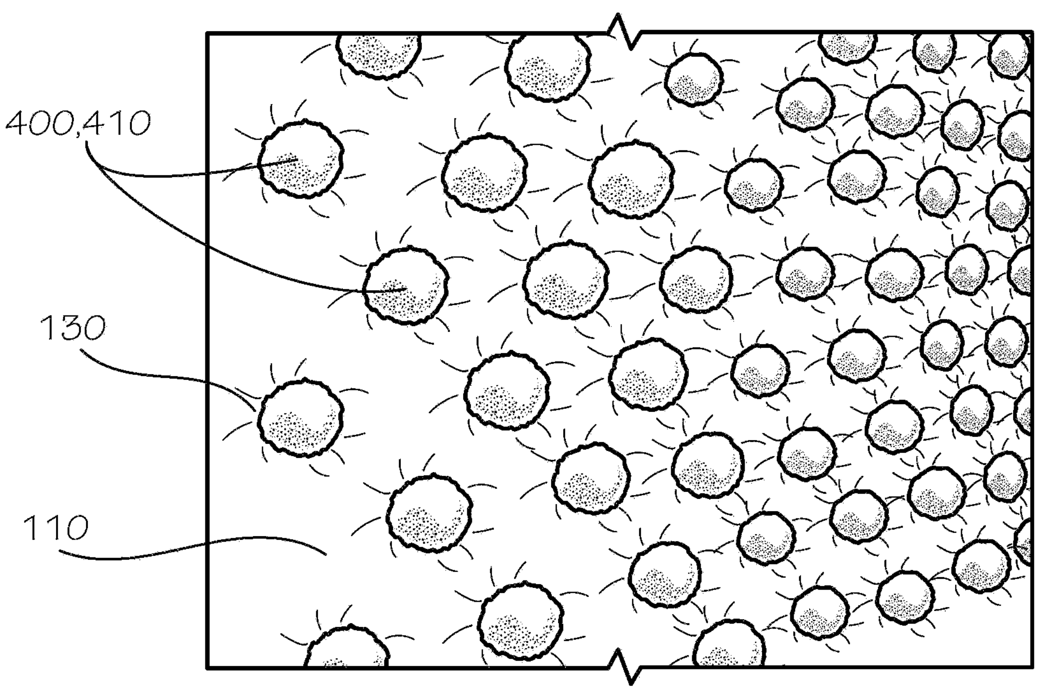


FIG. 12



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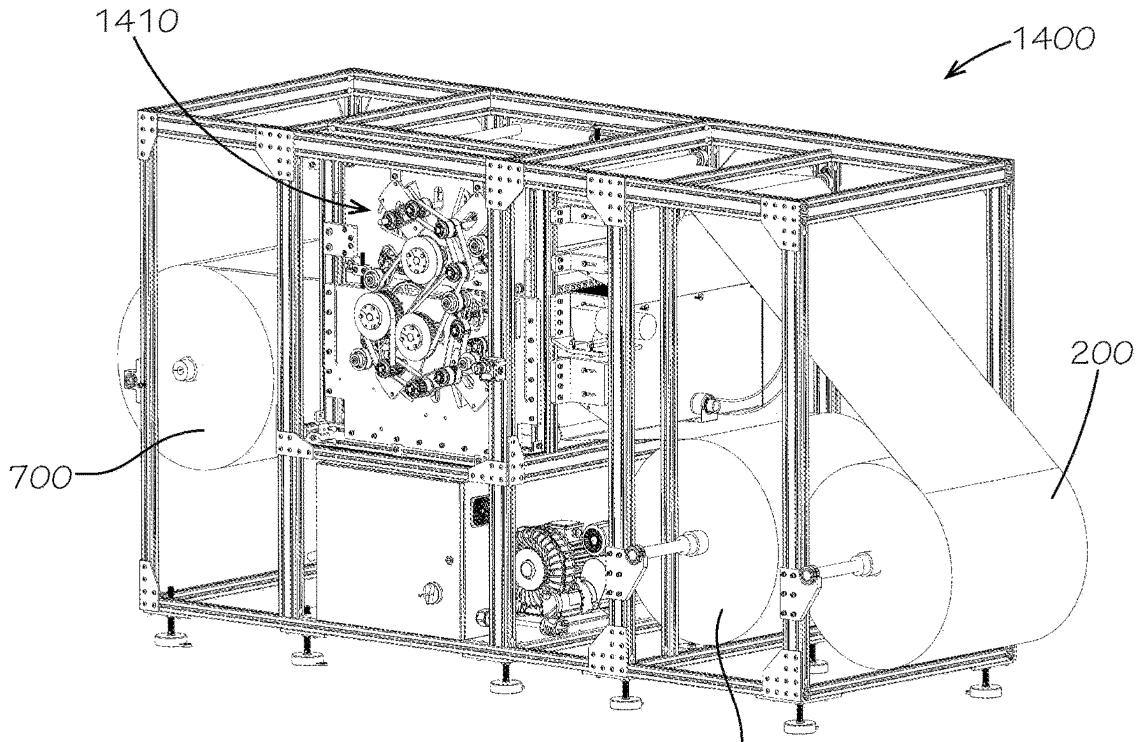
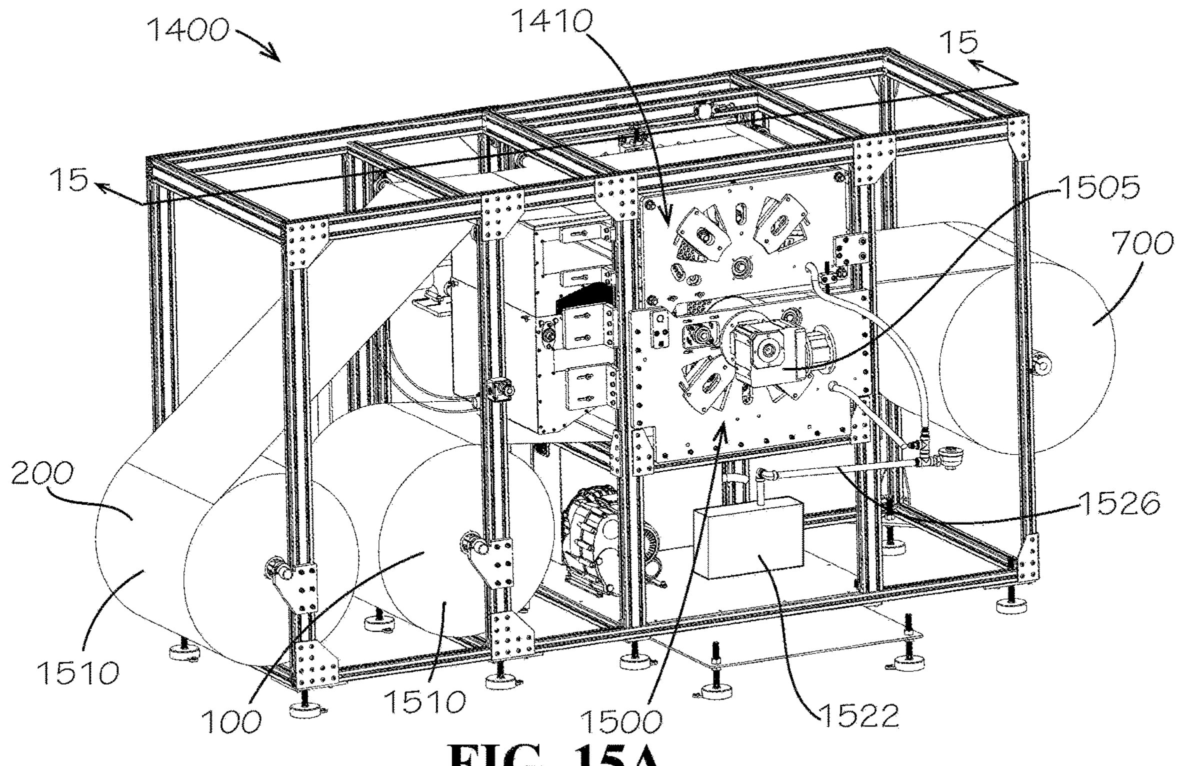


FIG. 14



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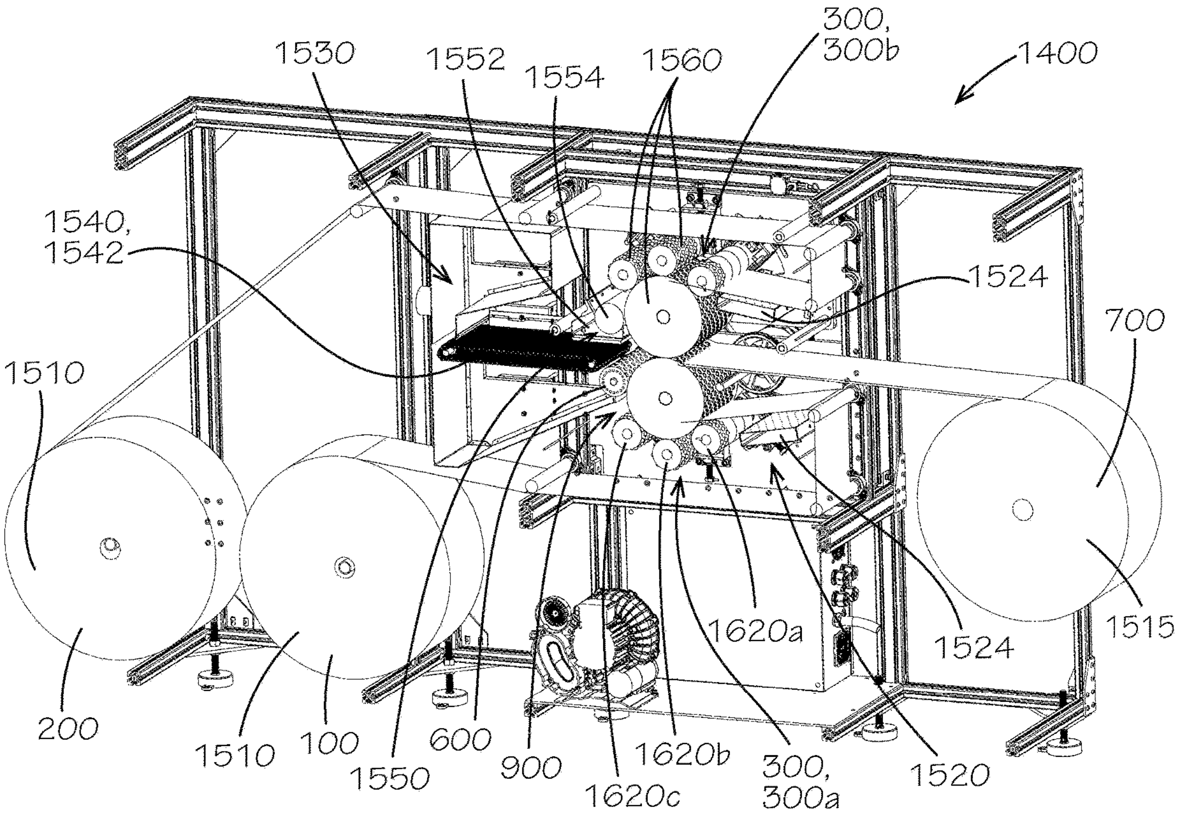


FIG. 15B

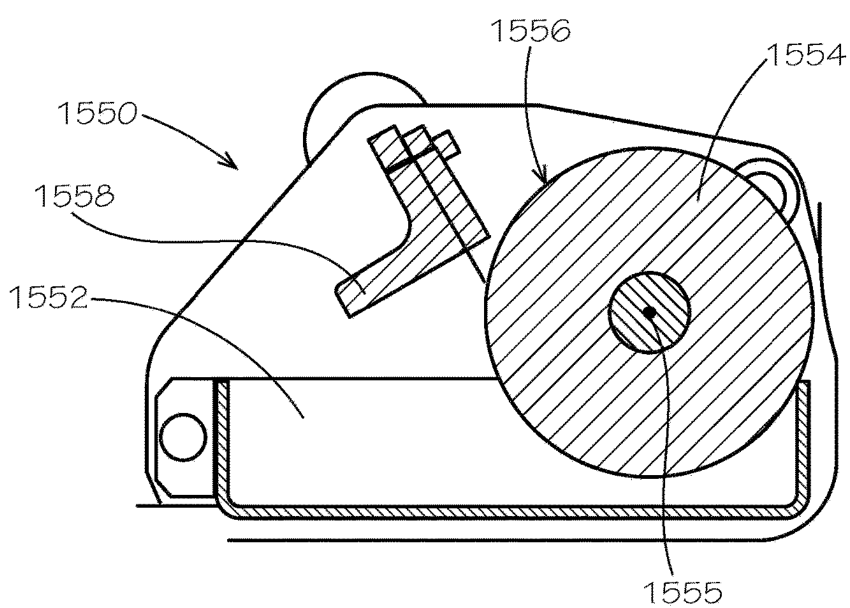
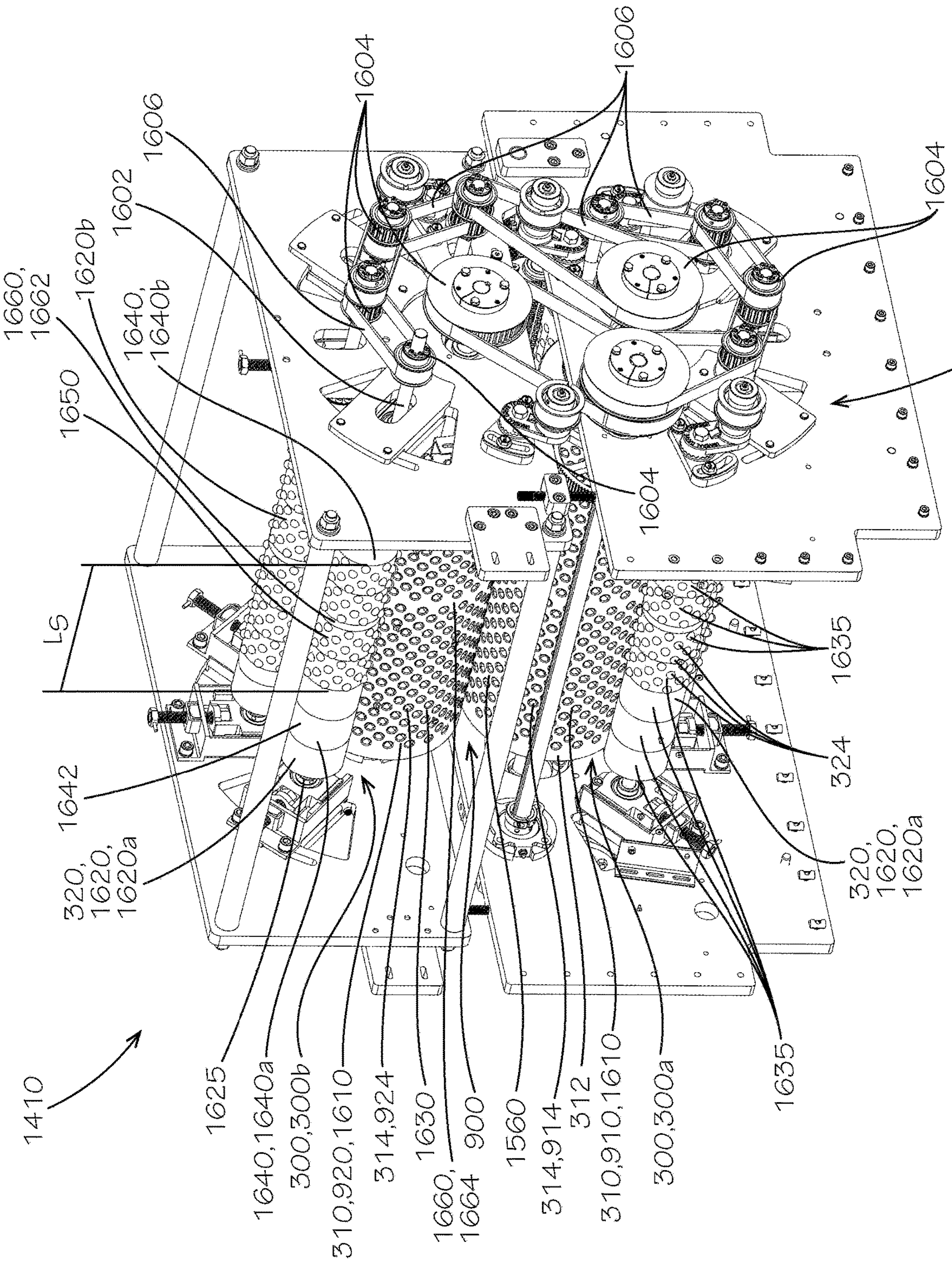


FIG. 15C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1620a
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FIG. 16

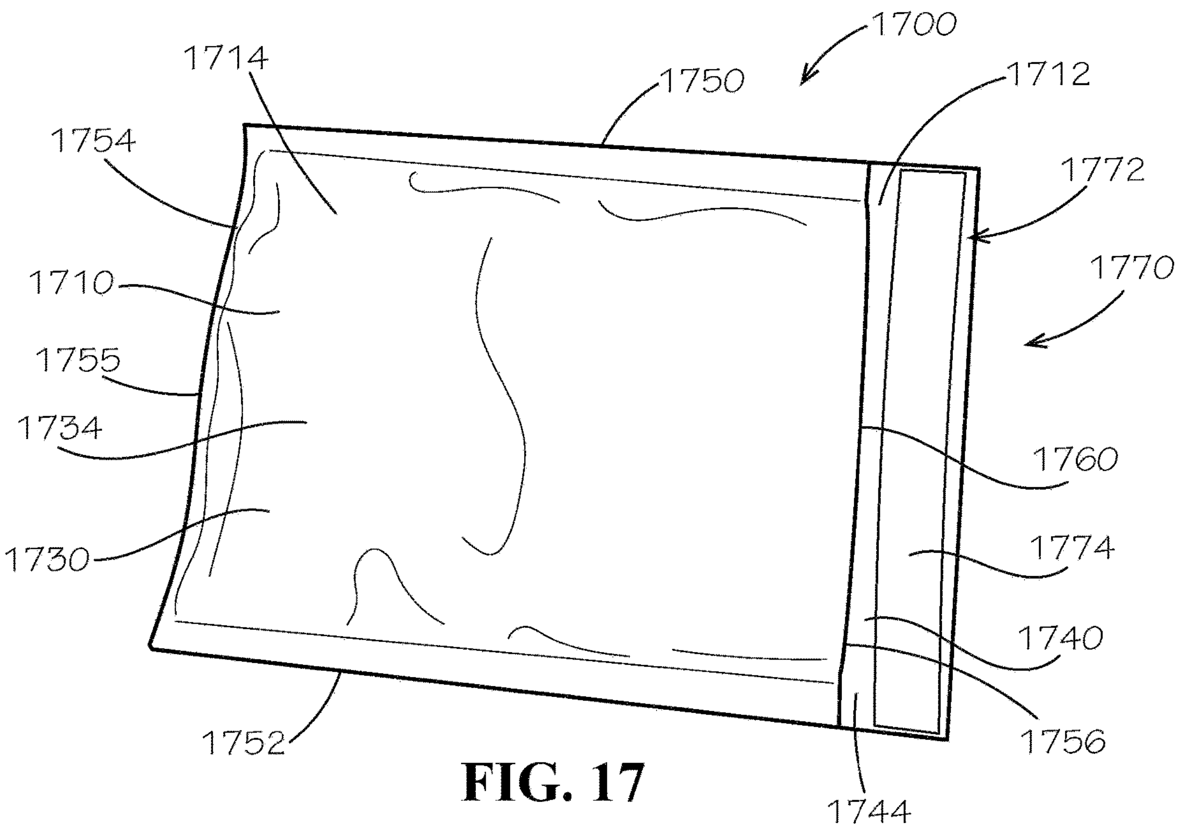
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FIG. 17

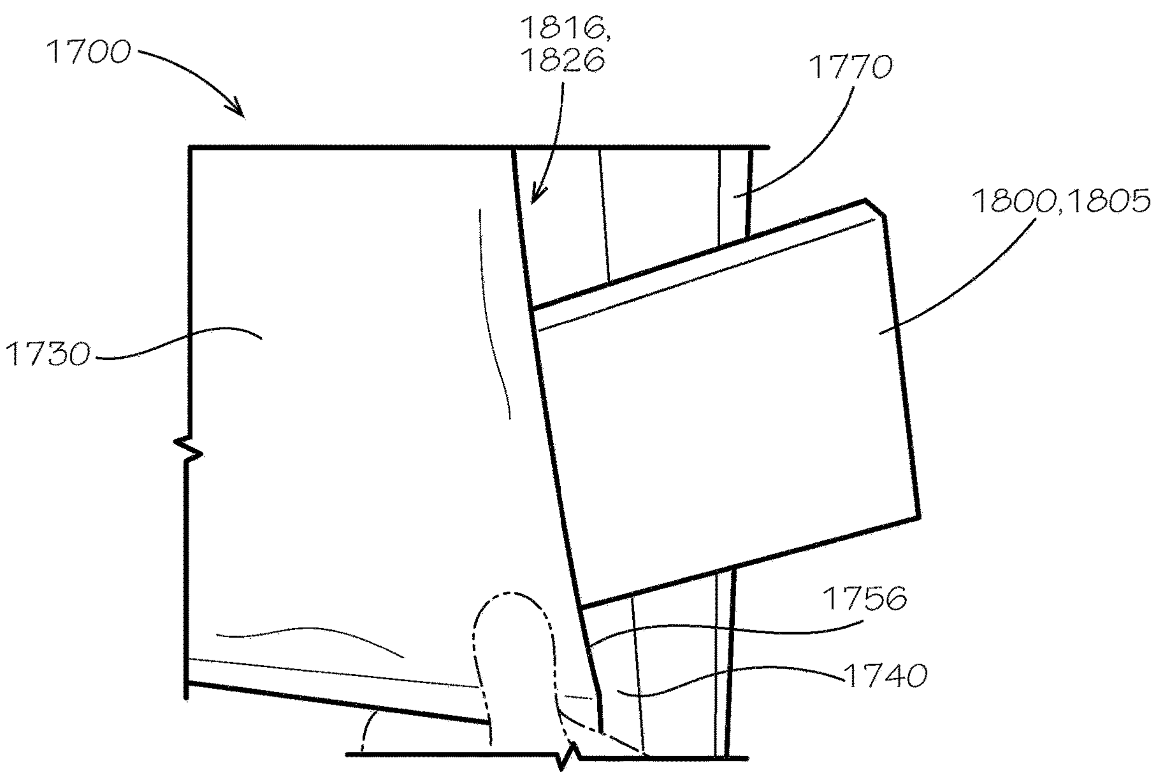
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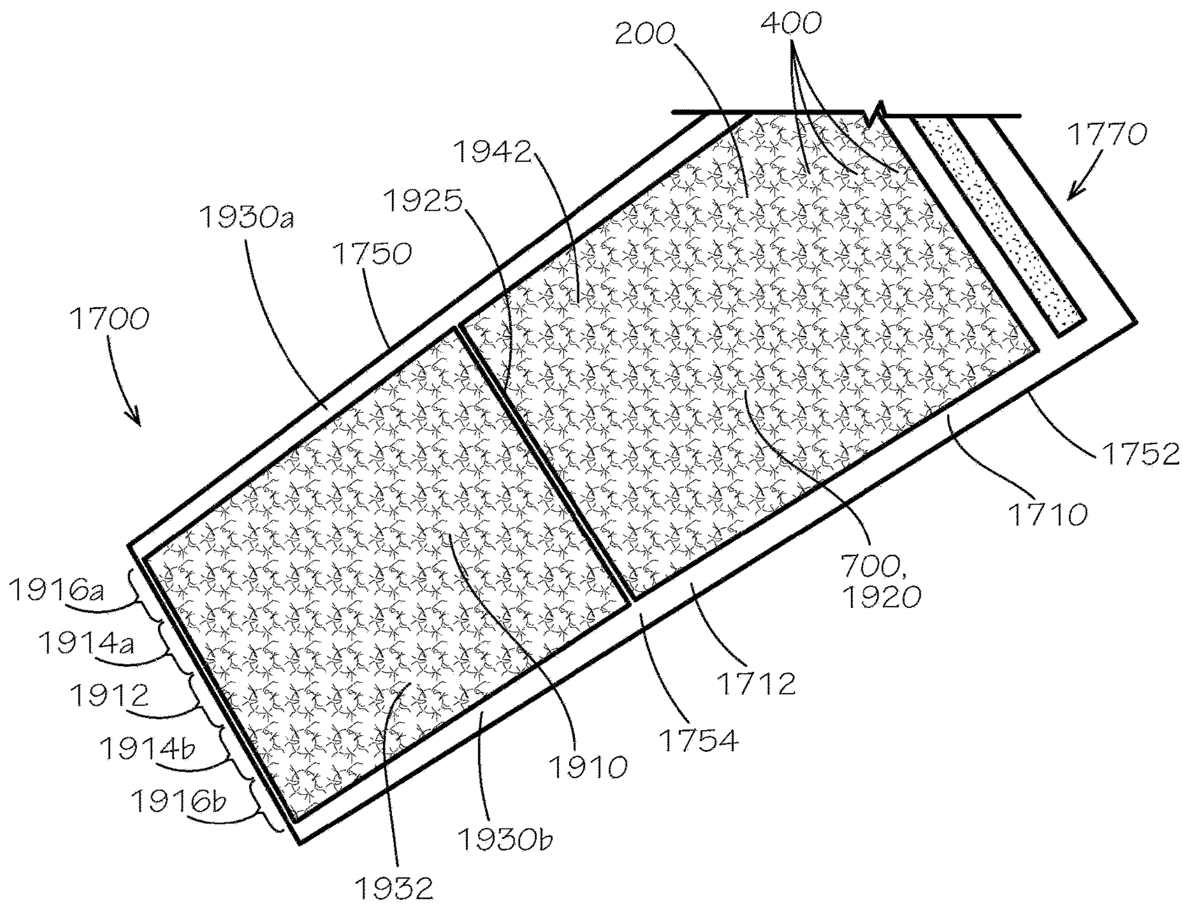
FIG. 18



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700,
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1916a
1914a
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1914b
1916b
1712
1754
1910
1930b
1932

FIG. 19

SHEET WITH CUSHIONING INSERTS

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application claims the benefit of U.S. Provisional Application No. 63/059,001, filed Jul. 30, 2020, which is hereby specifically incorporated by reference herein in its entirety

TECHNICAL FIELD

This disclosure relates to cushioning materials. More specifically, this disclosure relates to a sheet with cushioning inserts.

BACKGROUND

Cushioning sheets, such as wrapping paper and bubble wrap, are often used as cushioning for fragile objects during shipping. Wrapping paper typically comprises a single, thin sheet of paper that can be bunched up and packed around the object to limit movement of the object within a package and to provide cushioning from heavy impacts. However, wrapping paper easily crumples under force, and, once crumpled, can lose its volume and effectiveness. Bubble wrap typically comprises a plastic sheet defining air fill pockets. However, typical plastic bubble wrap is not recyclable.

SUMMARY

It is to be understood that this summary is not an extensive overview of the disclosure. This summary is exemplary and not restrictive, and it is intended neither to identify key or critical elements of the disclosure nor delineate the scope thereof. The sole purpose of this summary is to explain and exemplify certain concepts of the disclosure as an introduction to the following complete and extensive detailed description.

Disclosed is a cushioned sheet comprising a first layer comprising a first base and a plurality of first sockets extending from the first base, each of the first sockets spaced apart from adjacent ones of the first sockets; a second layer coupled to the first layer by an adhesive, the second layer comprising a second base and a plurality of second sockets extending from the second base, each of the second sockets aligned with a corresponding one of the first sockets to define a void therebetween; and a plurality of cushioning inserts, each of the plurality cushioning inserts substantially spherical in shape and received in a corresponding one of the voids.

Also disclosed is a cushioned mailer comprising a first layer defining a plurality of first sockets; a second layer defining a plurality of second sockets, each of the first sockets aligned with a corresponding one of the second sockets to define a void therebetween; and a cushioning insert received within each of the voids; wherein the first layer is coupled to the second layer to define a cushioning sheet, the cushioning sheet defining an inner cavity configured to receive contents, the inner cavity at least partially surrounded by the first layer, the second layer facing away from the inner cavity.

Additionally, a method of forming a cushioning sheet is disclosed, the method comprising forming a plurality of first sockets in a first layer; forming a plurality of second sockets in a second layer, each of the second sockets corresponding to a one of the first sockets; positioning a cushioning insert within a void defined between each corresponding pair of the first and second sockets; and coupling the first layer to the second layer.

Various implementations described in the present disclosure may include additional systems, methods, features, and advantages, which may not necessarily be expressly disclosed herein but will be apparent to one of ordinary skill in the art upon examination of the following detailed description and accompanying drawings. It is intended that all such systems, methods, features, and advantages be included within the present disclosure and protected by the accompanying claims.

BRIEF DESCRIPTION OF THE DRAWINGS

The features and components of the following figures are illustrated to emphasize the general principles of the present disclosure. Corresponding features and components throughout the figures may be designated by matching reference characters for the sake of consistency and clarity.

FIG. **1** is a top perspective view of a first paper layer, in accordance with one aspect of the present disclosure.

FIG. **2** is a bottom perspective view of a second paper layer, in accordance with another aspect of the present disclosure.

FIG. **3** is a top perspective view of a layer rolling device or forming the first paper layer of FIG. **1** and the second paper layer of FIG. **2**.

FIG. **4** is a top view of a plurality of spherical cushioning inserts, in accordance with another example aspect of the present disclosure.

FIG. **5** is a top perspective view of the spherical cushioning inserts of FIG. **4** applied to the first paper layer of FIG. **1**.

FIG. **6** is a perspective view of a vacuum roller configured to deposit the spherical cushioning inserts of FIG. **4** onto the first paper layer of FIG. **1**.

FIG. **7** is a top perspective view of the second paper layer of FIG. **2** laid over the first paper layer **1** of FIG. **1**, such that the spherical cushioning inserts of FIG. **4** are received between the first and second paper layers. The first paper layer, second paper layer, and spherical cushioning inserts can together define a cushioning sheet, in accordance with another aspect of the present disclosure.

FIG. **8** is a detail cross-sectional view of the cushioning sheet of FIG. **7** taken along line **8-8** in FIG. **7**.

FIG. **9** is a perspective view of sheet rolling device configured to assist in adhering the first paper layer of FIG. **1** to the second paper layer of FIG. **2**, in accordance with another aspect of the present disclosure.

FIG. **10** is a top view of the cushioning sheet of FIG. **7** after going through the sheet rolling device of FIG. **9**.

FIG. **11** is a bottom view of the cushioning sheet of FIG. **7** after going through the sheet rolling device of FIG. **9**.

FIG. **12** is a top view of the spherical cushioning inserts, in accordance with another example aspect of the present disclosure.

FIG. **13** is a top perspective view of the spherical cushioning inserts of FIG. **12** applied to the first paper layer, in accordance with another example aspect of the present disclosure.

FIG. **14** is rear perspective view of a rolling machine for forming the cushioning sheet of FIG. **7**, in accordance with another example aspect of the present disclosure.

FIG. **15A** is a front perspective view of the rolling machine of FIG. **14**.

FIG. **15**B is a cross-sectional view of the rolling machine of FIG. **14** taken along line **15-15** in FIG. **15**A.

FIG. **15**C is a front view of an adhesive applicator of the rolling machine of FIG. **14**.

FIG. **16** is a detail perspective view of a roller assembly of the rolling machine of FIG. **14**.

FIG. **17** is a front view of a cushioned mailer comprising the cushioning sheet of FIG. **7**, in accordance with another example aspect of the present disclosure.

FIG. **18** is a front view of contents being inserted into an inner cavity of the cushioned mailer of FIG. **17**.

FIG. **19** is a front view of the cushioned mailer of FIG. **17** in an unfolded configuration.

DETAILED DESCRIPTION

The present disclosure can be understood more readily by reference to the following detailed description, examples, drawings, and claims, and the previous and following description. However, before the present devices, systems, and/or methods are disclosed and described, it is to be understood that this disclosure is not limited to the specific devices, systems, and/or methods disclosed unless otherwise specified, and, as such, can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting.

The following description is provided as an enabling teaching of the present devices, systems, and/or methods in its best, currently known aspect. To this end, those skilled in the relevant art will recognize and appreciate that many changes can be made to the various aspects of the present devices, systems, and/or methods described herein, while still obtaining the beneficial results of the present disclosure. It will also be apparent that some of the desired benefits of the present disclosure can be obtained by selecting some of the features of the present disclosure without utilizing other features. Accordingly, those who work in the art will recognize that many modifications and adaptations to the present disclosure are possible and can even be desirable in certain circumstances and are a part of the present disclosure. Thus, the following description is provided as illustrative of the principles of the present disclosure and not in limitation thereof.

As used throughout, the singular forms "a," "an" and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to "an element" can include two or more such elements unless the context indicates otherwise.

Ranges can be expressed herein as from "about" one particular value, and/or to "about" another particular value. When such a range is expressed, another aspect includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent "about," it will be understood that the particular value forms another aspect. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

For purposes of the current disclosure, a material property or dimension measuring about X or substantially X on a particular measurement scale measures within a range between X plus an industry-standard upper tolerance for the specified measurement and X minus an industry-standard lower tolerance for the specified measurement. Because tolerances can vary between different materials, processes and between different models, the tolerance for a particular measurement of a particular component can fall within a range of tolerances.

As used herein, the terms "optional" or "optionally" mean that the subsequently described event or circumstance can or cannot occur, and that the description includes instances where said event or circumstance occurs and instances where it does not.

The word "or" as used herein means any one member of a particular list and also includes any combination of members of that list. Further, one should note that conditional language, such as, among others, "can," "could," "might," or "may," unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain aspects include, while other aspects do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more particular aspects or that one or more particular aspects necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular aspect.

Disclosed are components that can be used to perform the disclosed methods and systems. These and other components are disclosed herein, and it is understood that when combinations, subsets, interactions, groups, etc. of these components are disclosed that while specific reference of each various individual and collective combinations and permutations of these may not be explicitly disclosed, each is specifically contemplated and described herein, for all methods and systems. This applies to all aspects of this application including, but not limited to, steps in disclosed methods. Thus, if there are a variety of additional steps that can be performed it is understood that each of these additional steps can be performed with any specific aspect or combination of aspects of the disclosed methods.

Disclosed is a cushioning sheet and associated methods, systems, devices, and various apparatus. Example aspects of the cushioning sheet can comprise a first layer, a second layer, and a plurality of cushioning inserts received between the first layer and the second layer. In some aspects, the cushioning inserts can be substantially spherical in shape and can comprise a starch material. It would be understood by one of skill in the art that the cushioning sheet is described in but a few exemplary embodiments among many. No particular terminology or description should be considered limiting on the disclosure or the scope of any claims issuing therefrom.

FIG. **1** is a top perspective view of a first layer **100** of a cushioning sheet **700** (shown in FIG. **7**), in accordance with one aspect of the present disclosure. According to example aspects, the cushioning sheet **700** can be formed by a rolling machine **1400** (shown in FIG. **14**). In the present aspect, the first layer **100** can comprise a paper material and can therefore be a first paper layer **110**. In some aspects, the paper material can be a tissue-grade paper material, which may be covered with a coating to increase strength while maintaining flexibility. In other aspects, the first layer **100** can comprise any other suitable material known in the art. As shown, according to example aspects, the first paper layer **110** can comprise a substantially planar first base **120** defining a first upper surface **122** and a first lower surface **824** (shown in FIG. **8**). The first paper layer **110** can further define a plurality of first sockets **130** extending substantially downward from the first lower surface **824** of the planar first base **120**, relative to the orientation shown. In the present aspect, each of the first sockets **130** can be formed as a substantially semi-spherical dome and can define a substantially semi-spherical first recess **132**. As shown, each of the first sockets **130** are substantially uniform in size and shape, and each of the first recesses **132** are substantially uniform in size and shape. Each of the first sockets **130** can further define a substantially circular first opening **134** formed at the planar first base **120** that can allow access to the corresponding semi-spherical first recess **132**. In the present aspect, each of the first sockets **130** can be spaced apart from adjacent first sockets **130**, and the first sockets **130** can be oriented in a plurality of linear rows and columns; however, in other aspects, the first sockets **130** can be oriented in any other suitable arrangement or pattern.

FIG. **2** illustrates a second layer **200** of the cushioning sheet **700** (shown in FIG. **7**), in accordance with an aspect of the present disclosure. Like the first layer **100** (shown in FIG. **1**), the second layer **200** can comprise a paper material and can therefore be a second paper layer **210**. However, in other aspects, the second layer **200** can comprise any other suitable material known in the art, and may or may not comprise the same material as the first layer **100**. Example aspects of the second paper layer **210** can be substantially similar as the first paper layer **110** (shown in FIG. **1**), as described above. For example, the second paper layer **210** can comprise a substantially planar second base **220** defining a second upper surface **722** (shown in FIG. **7**) and a second lower surface **224**. The second paper layer **210** can further comprise a plurality of second sockets **230** extending from the planar second base **220**. However, unlike the first paper layer **110**, wherein the first sockets **130** (shown in FIG. **1**) can extend substantially downward from the first lower surface **824** (shown in FIG. **1**), the second sockets **230** can extend substantially upward from the second upper surface **722** of the second base **220**, relative to the orientation shown. In the present aspect, each of the second sockets **230** can be formed as a substantially semi-spherical dome and can define a substantially semi-spherical second recess **232**. As shown, each of the second sockets **230** are substantially uniform in size and shape, and each of the second recesses **232** are substantially uniform in size and shape. Each of the second sockets **230** can further define a substantially circular second opening **234** formed at the planar second base **220** that can allow access to the corresponding semi-spherical second recess **232**. In the present aspect, each of the second sockets **230** can be spaced apart from adjacent second sockets **230**, and the second sockets **230** can be oriented in a plurality of linear rows and columns; however, in other aspects, the second sockets **230** can be oriented in any other suitable arrangement or pattern. According to example aspects, the second sockets **230** and second recesses **232** of the second paper layer **210** can be substantially similar in size, shape, and arrangement to the first sockets **130** and first recesses **132** (shown in FIG. **1**) of the first paper layer **110**.

In some example aspects, as shown, an adhesive **240** may be applied to the second lower surface **224** of the planar second base **220**. The adhesive **240** can be configured to secure the second layer **200** to the first layer **100**, as described in further detail below. In other aspects, the adhesive **240** may instead be applied to the first upper surface **122** (shown in FIG. **1**) of the planar first base **120** (shown in FIG. **1**). In the present aspect, a thin film or layer of the adhesive **240** can be applied to the second lower surface **224**; in other aspects, however, a substantially thick film or layer of the adhesive **240** may be applied. According to example aspects, the adhesive **240** can comprise a starch material, which can be repulpable in some aspects. For example, the adhesive **240** can be a thin film of a starch paste **242**. In other aspects, the adhesive **240** can be any other suitable adhesive, including but not limited to, plastic adhesives, tape, and the like. As shown, in the present aspect, the adhesive **240** is applied to the second lower surface **224** only though in other aspects, the adhesive **240** may also be applied within the second sockets **230**. Furthermore, while the adhesive **240** is depicted herein as covering only select portions of the second lower surface **224** of the planar second base **220**, in other aspects, the adhesive **240** may substantially or entirely cover the second lower surface **224**.

FIG. **3** illustrates a layer rolling device **300** for forming the first and second sockets **130**,**230** of the first and second layers **100**,**200** (shown in FIGS. **1** and **2**, respectively). As shown, the layer rolling device **300** can comprise a first layer roller **310** and a second layer roller **320**. Each of the first and second layer rollers **310**,**320** can be substantially cylindrical in shape and can define a cylindrical outer surface **312**,**322**, respectively. The first layer roller **310** can define a plurality of semi-spherical socket indentations **314** formed in the corresponding cylindrical outer surface **312**, each of which can be substantially uniform in size and shape. The second layer roller **320** can define a plurality of semi-spherical socket projections **324** extending from the corresponding cylindrical outer surface **322**, each of which can be substantially uniform in size and shape, and which can be configured to fit within a corresponding one of the socket indentations **314**. According to example aspects, the first layer roller **310** and second layer roller **320** can be configured to rotate in unison, as each of the first layer **100** and second layer **200** are fed through the layer rolling device **300** between the first and second layer rollers **310**,**320**. Prior to passing through the layer rolling device **300**, each of the first and second layers **100**,**200** can be substantially planar. In some aspects, the first and second sockets **130**,**230** of the first and second layers **100**,**200** can be formed simultaneously by feeding the first and second layers **100**,**200** through the layer rolling device **300** at the same time, one on top of the other. However, in other aspects, such as the present aspect, the first layer **100** can be formed first, followed by the second layer **200**, or vice versa.

In example aspects, each of the socket indentations **314** of the first layer roller **310** can be configured to align with a corresponding one of the socket projections **324** of the second layer roller **320** as the first and second layer rollers **310**,**320** confront one another during rolling. Each of the socket projections **324** can be configured to push a corresponding portion of the first layer **100** and second layer **200** into the corresponding one of the socket indentations **314** to form the first sockets **130** and second sockets **230**, respectively. As such, each of the socket indentations **314** can be sized and shaped about equal to the first and second sockets **130**,**230**, and each of the socket projections **324** can be sized and shaped about equal to the first and second recesses **132**,**232** (shown in FIGS. **1** and **2**, respectively). Furthermore, as the first and second layers **100**,**200** are fed through the layer rolling device **300**, the portions of the cylindrical outer surfaces **312**,**322** of the first and second layer rollers **310**,**320** extending between the socket indentations **314** can press the planar first base **120** and planar second base **220** (shown in FIGS. **1** and **2**, respectively) therebetween to maintain the flat, planar shape thereof.

FIG. **4** illustrates a plurality of cushioning inserts **400** configured to be received between the first and second layers **100**,**200** (shown in FIGS. **1** and **2**, respectively), as described in further detail below. In the present aspect, each of the cushioning inserts **400** can define a substantially spherical shape, as shown. Moreover, in the present aspect, each of the cushioning inserts **400** can comprise a starch material, such that the cushioning inserts **400** can be starch cushioning inserts **410**. For example, the starch material can comprise corn starch in some aspects. In other aspects, the cushioning inserts **400** may define any other suitable shape and/or may comprise any other suitable starch material or other material known in the art. The cushioning inserts **410** can further be repulpable in some aspects. According to example aspects, the cushioning inserts **400** can be formed by an extrusion process, followed by a chopping process. In some example aspects, in the extrusion process, the corn starch material can be extruded through a chamber with a small amount of moisture. The moisture can vaporize under heat during the extrusion, and an elongated, continuous tube of corn starch material can be formed. In the chopping process, a cutter can cut the tube of corn starch material into short segments, each of which can puff up and harden and cool into a substantially spherical shape. In the present aspect, each of the spherical cushioning inserts **400** can define a diameter D_1 of about 0.375 inches. However, in other aspects, the cushioning inserts **400** can define any other suitable size and shape and/or can be formed by any other suitable processes known in the art.

FIG. **5** illustrates a first step in assembling the cushioning sheet **700** of FIG. **7**. As shown, each of the spherical cushioning inserts **400** can be placed into a corresponding one of the semi-spherical first sockets **130** of the first paper layer **110**, such that each of the spherical cushioning inserts **400** is partially received within a corresponding one of the semi-spherical first recesses **132** and can rest on a bottom **532** of the first socket **130**. For example, a lower portion **502** of each spherical cushioning insert **400** can be received within the corresponding first recess **132**, while an upper portion **504** of the spherical cushioning insert **400** can extend past the corresponding first opening **134** to be oriented outside of the first recess **132**. In the present aspect, the spherical cushioning inserts **400** can be sized such that a clearance is provided between the each of the spherical cushioning inserts **400** and corresponding sides **534** of the semi-spherical first socket **130**, as shown. As such, each of the spherical cushioning inserts **400** can be free to roll around within the corresponding semi-spherical first recess **132**. In some aspects, however, the clearance defined between the spherical cushioning inserts **400** and the sides **534** of the corresponding semi-spherical first sockets **130** may be reduced such that movement of the cushioning inserts **400** within the corresponding semi-spherical first recesses **132** may be limited or prohibited. Furthermore, in some aspects, the spherical cushioning inserts **400** may be secured to the first paper layer **110** within the corresponding first recesses **132** to limit or prohibit movement therein by a fastener, such as, for example, an adhesive, such as glue. For example, the glue can be PVA (polyvinyl acetate) glue in some aspects.

According to example aspects, as shown in FIG. **6**, a vacuum roller **600** may be provided for picking up the spherical cushioning inserts **400** (shown in FIG. **4**) and depositing them onto the first paper layer **110** (shown in FIG. **1**). As shown, in the present aspect, the vacuum roller **600** can define a substantially cylindrical outer surface **602** and a plurality of insert indentations **604** formed therein, each of which can be configured to pick up and retain one of the spherical cushioning inserts **400** therein. For example, each of the insert indentations **604** may comprise a vacuum port **606** configured to draw a corresponding one of the cushioning inserts **400** into the insert indentation **604**. Each spherical cushioning insert **400** can then be released from the corresponding insert indention into a corresponding one of the first recesses **132** (shown in FIG. **1**) of the first paper layer **110**. In other aspects, the spherical cushioning inserts **400** can be deposited into the corresponding first recesses **132** by any other suitable method known in the art. In other aspects, the cushioning inserts **400** can be deposited by the vacuum roller **600** or any other suitable method into the second recesses **232** (shown in FIG. **2**) of the second paper layer **210** (shown in FIG. **2**), instead of the first paper layer **110**.

FIGS. **7** and **8** illustrate a second step in forming the cushioning sheet **700**, wherein the second layer **200** can be laid over the first layer **100** (shown in FIG. **8**), or vice versa, with the spherical cushioning inserts **400** (shown in FIG. **8**) received therebetween. As such, in the present aspect, the first layer **100** can be a lower layer and the second layer **200** can be an upper layer, relative to the orientations shown. Referring to FIG. **8**, the first layer **100**, second layer **200**, and spherical cushioning inserts **400** can together define the cushioning sheet **700**. In the present aspect, the first layer **100** can be the first paper layer **110**, the second layer **200** can be the second paper layer **210**, and the cushioning sheet **700** can be wrapping paper **800** within which an object can be wrapped for cushioned protection. As shown, the second paper layer **210** can be positioned over the first paper layer **110** such that each of the second sockets **230** of the second paper layer **210** can be aligned with a corresponding one of the first sockets **130** of the first paper layer **110**. Each corresponding pair of first and second sockets **130**,**230** can together define a hollow, substantially spherical void **830** or bubble.

The spherical void **830** can comprise the corresponding semi-spherical first recess **132** and semi-spherical second recess **232**, and each of the cushioning inserts **400** can reside between a corresponding pair of first and second sockets **130**,**230** within the spherical void **830**. As shown, the lower portion **502** of the cushioning insert **400** can generally reside within the corresponding first recess **132** and the upper portion **504** of the cushioning insert **400** can generally reside within the corresponding second recess **232**. In the present aspect, a diameter of each void **830** can be larger than the diameter D_1 (shown in FIG. **4**) of the corresponding cushioning insert **400** and each of the cushioning inserts **400** can be received loosely within the void **830**, allowing the cushioning insert **400** to move around within the void **830**. In other aspects, the size of the voids **830** may be about equal to the size of the cushioning inserts **400** and/or the cushioning inserts **400** may be secured within the void **830** to limit or prohibit movement therein. Other aspects of the cushioning sheet **700** may not comprise the cushioning inserts **400**, and the voids **830** can be filled with any other suitable cushioning material. In other aspects, the voids **830** can be filled with air (or any other suitable gas), such that a plurality of cushioning air pockets can be formed between the first and second layers **100**,**200**. In such aspects, the first and second layers **100**,**200** may comprise a specific paper material that can be configured to limit or prevent the air from leaking out of the voids **830**. In some aspects, the first and second layers **100**,**200** can comprise a stretchy paper material. In some aspects, the first and second layers **100**, **200** can be coated in a material configured to limit or prevent the passage of air therethrough. Other aspects of the first and second layers **100** can comprise any suitable material, including paper and non-paper materials, configured to limit or prevent the passage of air therethrough.

As described above, example aspects of the second paper layer **210** can comprise an adhesive **240** (shown in FIG. **2**) applied to the second lower surface **224** thereof. The second lower surface **224** of the second paper layer **210** can engage the first upper surface **122** of the first paper layer **110**, and the adhesive **240** can join the second lower surface **224** to the first upper surface **122**, thereby securing the first and second paper layers **110**,**210** together. In some aspects, the second step of applying the second paper layer **210** over the first paper layer **110** can be the final step in forming the cushioning sheet **700**. However, in other aspects, a third step can comprise passing the cushioning sheet **700** through a sheet rolling device **900** (shown in FIG. **9**). As described above, the adhesive **240** can be repulpable in some aspects. In some example aspects, additional components of the cushioning sheet **700**, or the entire cushioning sheet **700**, can comprise repulpable materials.

FIG. **9** illustrates the sheet rolling device **900**, which can be provided for pressing the planar first base **120** of the first paper layer **110** (both **110** and **120** shown in FIG. **1**) against the planar second base **220** of the second paper layer **210** (both **210** and **220** shown in FIG. **2**) to ensure that the adhesive **240** (e.g., the starch paste **242**, both **240** and **242** shown in FIG. **2**) secures the first and second paper layers **110**,**210** to one another. As shown, the sheet rolling device **900** can define a first sheet roller **910** and a second sheet roller **920**. Each of the first and second sheet rollers **910**,**920** can define a cylindrical outer surface **912**,**922**, respectively and a plurality of semi-spherical indentations **914**,**924**, respectively, which can be sized about equal to or greater than each of the first and second sockets **130**,**230**. The first and second sheet rollers **910**,**920** can be configured to rotate in unison, as the cushioning sheet **700** (e.g., the wrapping paper **800**), both shown in FIG. **8**, is fed through the sheet rolling device **900** between the first and second sheet rollers **910**,**920**. According to example aspects, each of the indentations **914** of the first sheet roller **910** can be configured to align with a corresponding one of the indentations **924** of the second sheet roller **920** as the first and second sheet rollers **910**,**920** confront one another during rolling. As the cushioning sheet **700** is fed through the sheet rolling device **900**, each of the first sockets **130** (shown in FIG. **1**) of the first paper layer **110** can align with and extend into a corresponding one of the indentations **914** of the first sheet roller **910**, and similarly, each of the second sockets **230** (shown in FIG. **2**) of the second paper layer **210** can align with and extend into a corresponding one of the indentations **924** of the second sheet roller **920**. In this way, each of the semi-spherical first and second sockets **130**,**230**, as well as the spherical voids **830** (shown in FIG. **8**) formed by the first and second sockets **130**,**230**, can maintain their shapes as the cushioning sheet **700** is passed through the sheet rolling device **900**.

Furthermore, as the cushioning sheet **700** is fed through the sheet rolling device **900**, the portions of cylindrical outer surfaces **912**,**922** of the first and second sheet rollers **910**, **920** extending between the indentations **914**,**924** can be configured to contact the planar first and second bases **120**,**220** of the first and second paper layers **110**,**210**, respectively. The first and second sheet rollers **910**,**920** can press the first upper surface **122** (shown in FIG. **1**) of the first paper layer **110** against the second lower surface **224** (shown in FIG. **2**) of the second paper layer **210**, such that the adhesive **240** applied to the second lower surface **224** firmly engages the first upper surface **122**, thereby further securing the first and second paper layers **110**,**210** together. As described above, in some aspects, a substantially thin film or layer of the adhesive **240** can be applied to the second lower surface **224**. However, in some aspects, the film or layer of adhesive **240** can be thick enough such that some of the adhesive **240** may be pressed into some or all of the voids **830**, forming an adhesive ring around the corresponding cushioning insert **400** (shown in FIG. **4**) therein, which can aid in holding the cushioning insert **400** in place within the void **830** and/or which may increase the strength and/or cushioning ability of the cushioning sheet **700** in some aspects. FIG. **10** illustrates a top view of the cushioning sheet **700** after being passed through the sheet rolling device **900** (shown in FIG. **9**) to ensure the first base **120** (shown in FIG. **1**) of the first paper layer **110** (shown in FIG. **1**) is adequately secured to the second base **120** of the second paper layer **210**, and FIG. **11** illustrates a bottom view of the cushioning sheet **700** after passing through the sheet rolling device **900**.

FIG. **12** illustrates the cushioning inserts **400** according to another example aspect of the present disclosure, wherein the cushioning inserts **400** again define a substantially spherical shape. In the present aspect, the spherical shape of the cushioning inserts **400** is even more defined that the substantially spherical cushioning inserts shown in FIG. **4**. Furthermore, in the present aspect, the cushioning inserts **400** can be the starch cushioning inserts **410**. FIG. **13** illustrates each of the cushioning inserts **400** of FIG. **12** placed into a corresponding one of the semi-spherical first sockets **130** of the first paper layer **110**.

FIGS. **14** and **15**A-B illustrate the rolling machine **1400**, according to an example aspect of the present disclosure, for forming the cushioning sheet **700**. FIG. **14** illustrates a rear perspective view of the rolling machine **1400**, FIG. **15**A illustrates a front perspective view of the rolling machine **1400**, and FIG. **15**B illustrates a cross-sectional view of the rolling machine **1400** taken along line **15-15** in FIG. **15**A. Example aspects of the rolling machine **1400** can comprise a roller assembly **1410**, which can comprise a plurality of rollers **1560** (shown in FIG. **15**B). In the present aspect, the roller assembly **1410** can comprise two of the layer rolling devices **300** (shown in FIG. **15**B) for simultaneously forming the first and second sockets **130**,**230** (shown in FIGS. **1** and **2**, respectively) in the first and second layers **100**,**200**, respectively. Other aspects of the rolling machine **1400** can comprise only one of the layer rolling devices **300**, which can form the first and second sockets **130**,**230** in the first and second layers **100**,**200** either simultaneously or consecutively. The roller assembly **1410** of the rolling machine **1400** can further comprise the sheet rolling device **900** (shown in FIG. **15**B) for sealing the first layer **100** to the second layer **200** with the cushioning inserts **400** (shown in FIG. **4**) received therebetween. Referring to FIG. **15**A, a drive system **1500** can be provided for driving the movement of the roller assembly **1410** during production of the cushioning sheet **700**. As shown, example aspects of the drive system **1500** can comprise one or more motors **1505** configured to impart rotational movement to the various rollers **1560** (shown in FIG. **15**B) of the roller assembly **1410**, as described in further detail below.

According to example aspects, each of the first and second layers **100**,**200** can be formed as a roll of raw material **1510** (e.g., a roll of paper, as shown) prior to being passed through the rolling machine **1400**. Referring to FIG. **15**B, the rolling machine **1400** can comprise the two layer rolling devices **300**—for example, a first layer rolling device **300***a* and a second layer rolling device **300***b*. In some aspects, the rolling machine **1400** can comprise one or more steam units **1520** configured to steam the first and second layers **100**,**200** prior to passage through the corresponding layer rolling devices **300**. Steaming the first and second layers **100**,**200** can soften the paper material, which can allow the paper material to stretch without tearing during formation of the first and second sockets **130**,**230** (shown in FIGS. **1** and **2**, respectively). Example aspects of the steam unit **1520** can comprise a water tank **1522** (shown in FIG. **15**A), one or more steam pans **1524**, and one or more conduits **1526** (shown in FIG. **15**A) for transferring water from the water tank **1522** to the steam pans **1524**. The steam pans **1524** can be configured to heat the water received from the water tank **1522** to produce steam. In example aspects, as shown, each of the first and second layers **100**,**200** can pass over a corresponding one of the steam pans **1524** before being fed into the corresponding layer rolling device **300**. The layer rolling devices **300** can then form the first and second sockets **130**,**230** in the first and second layers **100**,**200**, as described above. In some aspects, one or both the steam units **1520** can be turned off or left unused when it is desired to cold set the first and/or second layers **100**,**200**. Other aspects of the rolling machine **1400** may not comprise the steam units **1520** or the steam units **1520** can be removed as desired. Furthermore, in example aspects, the layer rolling devices **300** can be used to form the first and second sockets **130**,**230** in the corresponding first and second layers **100**, **200**, and the first and second layers **100**,**200** can then be removed from the rolling machine **1400** and used as textured sheets of material (e.g., a textured paper sheets). In some aspects, only one of the layer rolling devices **300** may be used or provided for forming a textured sheet from the first layer **100** and/or second layer **200**.

Once the first sockets **130** have been formed in the first layer **100** by the corresponding first layer rolling device **300***a*, the cushioning inserts **400** (shown in FIG. **4**) can be positioned within the first sockets **130**. Example aspects of the rolling machine **1400** can comprise the vacuum roller **600** for picking up the cushioning inserts **400** from a hopper **1530** or other repository and depositing them onto the first upper surface **122** (shown in FIG. **1**) of the first layer **100**. Some aspects of the rolling machine **1400** can also comprise a vibration mechanism **1540** configured to vibrate each of cushioning inserts **400** into a corresponding one of the insert indentations **604** (shown in FIG. **6**) of the vacuum roller **600**. For example, in the present aspect, the vibration mechanism **1540** can comprise a vibrating conveyor belt **1542**. The cushioning inserts **400** can be fed from the hopper **1530** onto the vibrating conveyor belt **1542**, which can agitate the cushioning inserts **400** thereon. The agitated cushioning inserts **400** can be transported along the conveyor belt **1542** and can be deposited onto the vacuum roller **600**. The movement of the cushioning inserts **400** caused by the vibrating conveyor belt **1542** can aid in locating the cushioning inserts **400** within the insert indentations **604**. For example, the agitated cushioning inserts **400** can slide, roll, or otherwise move across the cylindrical outer surface **602** (shown in FIG. **6**) of the vacuum roller **600** until dropping into a corresponding one of the insert indentations **604**. The cushioning inserts **400** can be retained within the corresponding insert indentations **604** of the vacuum roller **600** by suction provided through the corresponding vacuum port **606** (shown in FIG. **6**).

Additionally, once the second sockets **230** have been formed in the second layer **200** by the corresponding second layer rolling device **300**b, the adhesive **240** (shown in FIG. **2**) can be applied to the second lower surface **224** (shown in FIG. **2**) of the second layer **200**. According to example aspects, the rolling machine **1400** can comprise an adhesive applicator **1550** configured to apply the adhesive **240** to the second lower surface **224**. In the present aspect, the adhesive applicator **1550** can comprise an adhesive pan **1552** and an adhesive roller **1554**. A pool of the adhesive **240** can be received in the adhesive pan **1552**, and the adhesive roller **1554** can be continuously dipped into the pool of the adhesive **240** as the adhesive roller **1554** rotates, thereby continuously applying the adhesive **240** to the adhesive roller **1554**. After dipping into the pool of the adhesive **240**, the adhesive roller **1554** can rotate into contact with the second layer **200** to continuously apply the adhesive **240** to the second lower surface **224** thereof. An example aspect of the adhesive applicator **1550** is shown and described in further detail below with respect to FIG. **15**C.

Once the cushioning inserts **400** have been deposited on the first layer **100** and the adhesive **240** has been applied to the second layer **200**, each of the first and second layers **100**,**200** can be simultaneously passed through the sheet rolling device **900** in facing contact with one another. As described above, when passing the first and second layers **100**,**200** through the sheet rolling device **900**, each of the first sockets **130** of the first layer **100** can be aligned with a corresponding one of the second sockets **230** of the second layer **200** to define the substantially spherical void **830** (shown in FIG. **8**) therebetween. Each of the cushioning inserts **400** can be positioned within a corresponding one of the voids **830**, and the sheet rolling device **900** can adhere the first layer **100** to the second layer **200** to retain the cushioning inserts **400** within the corresponding voids **830**. The cushioning sheet **700** can then exit the sheet rolling device **900**, and in the present aspect, the cushioning sheet **700** can be wrapped around itself to define a finished roll **1515** of the cushioning sheet **700**.

FIG. **15**C illustrates a front view of the adhesive applicator **1550**. As shown, the adhesive applicator **1550** can comprise the adhesive pan **1552** within which a pool of the adhesive **240** (shown in FIG. **2**) can be received. The adhesive roller **1554** can be configured to rotate about a roller axis **1555** (going into the page) proximate to the adhesive pan **1552**, such that an outer roller surface **1556** of the adhesive roller **1554** can be continuously rotated into the pool of glue. The adhesive roller **1554** can further be continuously rotated into contact with the second layer **200** (shown in FIG. **14**) to apply the adhesive **240** thereto. For example, as a first portion of the outer roller surface **1556** rotates out of the pool of the adhesive **240** with the adhesive **240** applied thereto, the first portion can rotate into contact with the second layer **200**. Specifically, the first portion of the adhesive roller **1554** can rotate into contact with the second lower surface **224** (shown in FIG. **2**) of the second layer **200** as the second layer **200** moves through the roller assembly **1410** (shown in FIG. **14**) towards the sheet rolling device **900** (shown in FIG. **15**B), depositing the adhesive **240** on the second lower surface **224**. Furthermore, in some aspects, the adhesive applicator **1550** can further comprise a doctor blade **1558** configured to remove any excess adhesive **240** from the adhesive roller **1554** after application of the adhesive **240** to the second layer **200**. For example, after the first portion of the adhesive roller **1554** has applied the adhesive to the second layer **200**, the first portion can be rotated into contact with the doctor blade **1558**, which can scrape the outer roller surface **1556** to remove any excess adhesive **240** thereon prior to re-dipping the first portion into the pool of adhesive **240**.

FIG. **16** illustrates a detail view of a roller assembly **1410** of the rolling machine **1400** (shown in FIG. **14**), in accordance with an example aspect of the present disclosure. The roller assembly **1410** can comprise the pair of the layer rolling devices **300** and the sheet rolling device **900**. The layer rolling devices **300** and the sheet rolling device **900** can be operably connected to one another by the drive system **1500**, such that the various rollers **1560** thereof can rotate in unison. As described above, the drive system **1500** can comprise the motor **1505** (shown in FIG. **15**A). Furthermore, in the present aspect, each of the rollers **1560** can comprise a rotating axle **1602**, and the motor **1505** can be configured to directly drive one of the rotating axles **1602**. For example, in the present aspect, the motor **1505** can directly drive the rotating axle **1602** of the first layer roller **310** of the first layer rolling device **300***a*. Example aspects of the drive system **1500** can further comprise a wheel **1604** or sprocket mounted on each of the rotating axles **1602**, and the wheels **1604** or sprockets can be interconnected by one or more straps **1606** or chains to transmit rotational motion between the rollers **1560**.

According to example aspects, each of the layer rolling devices **300** can comprise one or more of the first layer rollers **310** defining the socket indentations **314**, which can be considered female rollers **1610**. Each of the layer rolling devices **300** can further comprise one or more of the second layer rollers **320** defining the socket projections **324**, which can be considered male rollers **1620**. For example, in the present aspect, each layer rolling device **300** can comprise three small male rollers **1620***a,b,c* (**1620***c* shown in FIG. **15**B) oriented in series and configured to engage one large female roller **1610**. Other aspects of the layer rolling devices **300** can comprise more female rollers **1610** and/or more or fewer male rollers **1620**. The male rollers **1620** and female roller **1610** of each layer rolling device **300** can be interconnected by the drive system **1500** as described above and can rotate concurrently as the corresponding first and second layers **100**,**200** (shown in FIG. **14**) are fed therethrough. As described above, each of the socket projections **324** of the male rollers **1620** can be configured to align with a corresponding one of the socket indentations **314** of the corresponding female roller **1610** as the female and male rollers **1610**,**1620** confront one another during rotation. Each of the socket projections **324** can be configured to push a portion of the corresponding first layer **100** or second layer **200** into the corresponding socket indentation **314** to form the first and second sockets **130**,**230** (shown in FIGS. **1** and **2**), respectively.

In some aspects, some or all of the female and male rollers **1610**,**1620** can be formed from a singular cylindrical structure **1630** extending from a first end **1625** of the corresponding roller **1560** to an opposite second end (not shown) of the corresponding roller **1560**. A length of the roller can be defined between the first end **1625** and the opposite second end. In some aspects, some or all of the female and male rollers **1610**,**1620** can be formed from a plurality of cylindrical collars **1635** stacked laterally between the corresponding first end **1625** and the second end. Rollers **1560** comprising the collars **1635** can allow for selective adjustment of the length of the roller **1560** by adding or removing collars **1635**. Furthermore, according to example aspects, each of the collars **1635** can be either a smooth collar **1640** defining a substantially smooth outer collar surface **1642** or a socket collar **1650** defining a plurality of the socket projections **324** or the socket indentations **314**. For example, in the present aspect, each of the female rollers **1610** can define the singular cylindrical structure **1630**, while each of the male rollers **1620** can define a plurality of the collars **1635**. As shown, each of the male rollers **1620***a,b* can comprise one or more first smooth collars **1640***a* proximate to the corresponding first end **1625** and one or more second smooth collars **1640***b* proximate to the corresponding second end. Each of the male rollers **1620***a,b* can further comprise one or more of the socket collars **1650** defining the socket projections **324** oriented centrally between the first and second smooth collars **1640***a,b*. The male roller **1620***c* (shown in FIG. **15**B) can comprise the socket collars **1650** extending from the first end **1625** to the second end, and does not comprise any of the smooth collars **1640** in the present aspect. In other aspects, however, the male roller **1620***c* may comprise the smooth collars **1640**.

According to example aspects, any of the rollers **1560** and/or collars **1635** can be selectively removed and replaced with rollers **1560** and/or collars **1635** having varying patterns, varying sizes, and/or other varying features. In one example, the rollers **1560** and/or collars **1635** can be replaced with rollers **1560** and/or collars **1635** defining larger or smaller socket indentations **314** and/or socket projections **324**. In another example, the rollers **1560** and/or collars **1635** can be replaced with rollers **1560** and/or collars **1635** defining socket indentations **314** and/or socket projections **324** defining an alternative shape (e.g., cuboidal instead of semi-spherical). In another example, some or all of the smooth collars **1640** can be replaced with socket collars **1650** as desired, or vice versa.

According to example aspects, each of the female and male rollers **1610**,**1620** can be about equal in length. In other aspects, some or all of the female and male rollers **1610**, **1620** can define varying lengths. Additionally, as shown, each of the smooth collars **1640** can be about equal in size, and each of the socket collars **1650** can be about equal in size. In other aspects, the smooth collars **1640** and/or socket collars **1650** can define varying sizes. In the present aspect, the male rollers **1620***b* can define more of the socket collars **1650** and fewer of the smooth collars **1640** than the male rollers **1620***a*. Similarly, the male rollers **1620***c* (shown in FIG. **15**B) can define more of the socket collars **1650** and fewer of the smooth collars **1640** than the male rollers **1620***b*. As such, a length L_s of the stack of socket collars **1650** of the male rollers **1620***a* can be less than the length L_s of the stack of socket collars **1650** of the male rollers **1620***b*, and the length L_s of the stack of socket collars **1650** of the male rollers **1620***b* can be less than the length L_s of the stack of socket collars **1650** of the male rollers **1620***c*. As described above, each of the collars **1635** can be selectively removed and replaced, and thus, the length L_s can be selectively adjusted. Additionally, in some aspects, a height to which the socket projections **324** extend away from the corresponding socket collars **1650** can vary between some or all of the male rollers **1620***a,b,c*. For example, in some aspects, the socket projections **324** of the male roller **1620***a* can extend to a height that is less that a height of socket projections **324** of the male roller **1620***b*, and the socket projections **324** of the male roller **1620***b* can extend to a height that is less than a height of the socket projections **324** of the male roller **1620***c*. In other aspects, the socket projections **324** of each of the male rollers **1620***a,b,c* can be about equal in height.

According to example aspects, each of the first and second layers **100**,**200** (shown in FIG. **14**) can be fed through the corresponding layer rolling devices **300** in the direction from the male roller **1620***a* to the male roller **1620***c*. The centrally-oriented socket collars **1650** of the male roller **1620***a* can form a center set **1912** (shown in FIG. **19**) of the first and second sockets **130**,**230** at and around a center **1910** (shown in FIG. **19**) of the first and second layers **100**,**200**, respectively. The socket collars **1650** of the male roller **1620***b* can then form first and second intermediate sets **1914***a,b* (shown in FIG. **19**) of the first and second sockets **130**,**230** on either side of the center set **1912**. Finally, the socket collars **1650** of the male roller **1620***c* can form first and second outer sets **1916***a,b* (shown in FIG. **19**) of the first and second sockets **130**,**230** adjacent to the first and second intermediate sets **1914***a,b* and distal to the center set **1912**.

In some aspects, each of the male rollers **1620** can define one or more spacer areas **1660** extending about a circumference of the male roller **1620**. For example, in the present aspect, each of the male rollers **1620** can define one of the spacer areas **1660** oriented about centrally along the length of the male roller **1620**. In other aspects, each of the male rollers **1620** can define additional spacer areas **1660** and/or the spacer area(s) **1660** can be positioned anywhere along the length of the male roller **1620**. As shown, each of the spacer areas **1660** can be substantially smooth and does not define the socket projections **324**. In the present aspect, each of the spacer areas **1660** of the male rollers **1620** can be provided by a spacer collar **1662** positioned between adjacent socket collars **1650**. In some aspects, the spacer collar **1662** can be similar to the smooth collars **1640**. Furthermore, the spacer area **1660** of the male roller **1620***a* can be aligned with corresponding spacer areas **1660** of the male rollers **1620***b*,**1620***c*. According to example aspects, the spacer areas **1660** can define cutting portions extending lengthwise along the first and second layers **100**,**200**, which can be substantially planar and which do not define the corresponding first and second sockets **130**,**230**. The cutting portions of the first and second layers **100**,**200** can be aligned on the finished cushioning sheet **700**, such that the cushioning sheet **700** can be cut lengthwise along the cutting portions without piercing any of the cushioning inserts **400** or voids **830**. However, other aspects of the male rollers **1620** may not comprise the spacer areas **1660**.

In some aspects, as shown, each of the female rollers **1610** can also define one or more of the spacer areas **1660**. For example, in the present aspect, each of the female rollers **1610** can define one of the spacer areas **1660** oriented about centrally along a length of the female roller **1610**. Each of the spacer areas **1660** of the female rollers **1610** can be formed as a smooth circumferential region **1664** of the cylindrical outer surface **312** of the female roller **1610**. The smooth circumferential region **1664** of the female roller **1610** can be configured to align with the spacer collars **1662** of the male rollers **1620**. In other aspects, however, the female rollers **1610** may not comprise the corresponding spacer areas **1660**. The roller assembly **1410** can further comprise the sheet rolling device **900**. According to example aspects, the sheet rolling device **900** can comprise each of the female rollers **1610** of the layer rolling devices **300**. Thus, the female rollers **1610** of the layer rolling devices **300** can serve as the first and second sheet rollers **910**,**920**, and the corresponding socket indentations **314** of the female rollers **1610** can serve as the indentations **914**,**924**. The first and second sheet rollers **910**,**920** can confront another, and each of the first and second layers **100**,**200** can be simultaneously fed therebetween. As described above, each of the indentations **914** of the first sheet roller **910** can align with a corresponding one of the indentations **924** of the second sheet roller **920** as the first and second sheet rollers **910**,**920** are concurrently rotated. The first sockets **130** of the first layer **100** can extend into the indentations **914** of the first sheet roller **910**, and the second sockets **230** of the second layer **200** can extend into the indentations **924** of the second sheet roller **920**, allowing the voids **830** (shown in FIG. **8**) therebetween to maintain their shape and preventing the cushioning inserts **400** (shown in FIG. **4**) from being compressed or crushed therein. Additionally, as described above, the cylindrical outer surfaces **912**,**922** (shown in FIG. **9**) of the first and second sheet rollers **910**,**920** surrounding the corresponding indentations **914**,**924** can press the second lower surface **224** (shown in FIG. **2**) of the second layer **200** into contact with the first upper surface **122** (shown in FIG. **1**) of the first layer **100**. The adhesive **240** (shown in FIG. **2**) applied to the second lower surface **224** of the second layer **200** can adhere to the first upper surface **122** of the first layer **100** to secure the first and second layers **100**,**200** together with the cushioning inserts **400** received in the voids **830**, thereby defining the cushioning sheet **700** (shown in FIG. **7**). The cushioning sheet **700** can then exit the roller assembly **1410** and can be wrapped around itself to define the finished roll **1515** (shown in FIG. **15**B).

Thus, a method of forming the cushioning sheet **700** can comprise forming the plurality of first sockets **130** in the first layer **100**, and forming the plurality of second sockets **230** in the second layer **200**, wherein each of the second sockets **230** can correspond to a one of the first sockets **130**. The method can further comprise positioning one of the cushioning inserts **400** within the void **830** defined between each corresponding pair of the first and second sockets **130**,**230**, and coupling the first layer **100** to the second layer **200**.

FIGS. **17**-**19** illustrate a cushioned mailer **1700**, in accordance with an example aspect of the present disclosure. Referring to FIG. **17**, example aspects of the cushioned mailer **1700** can comprise an outer sheet **1710** defining an inner surface **1712** and an outer surface **1714**. The outer sheet **1710** can be a substantially planar paper sheet in the present aspect, though in other aspects, the outer sheet **1710** can comprise any other suitable material and/or configuration. According to example aspect, the inner surface **1712** of the outer sheet **1710** can define an outer cavity **1816** (shown in FIG. **18**). The cushioned mailer **1700** can further comprise the cushioning sheet **700** (shown in FIG. **19**), which can be an inner sheet **1920** (shown in FIG. **19**) in the present aspect, positioned within the outer cavity **1816** and coupled to the inner surface **1712** of the outer sheet **1710**. For example, the cushioning sheet **700** can be coupled to the inner surface **1712** by an adhesive, such as, for example, glue, or any other suitable fastener known in the art. Other aspects of the cushioning sheet **700** may not be coupled to the outer sheet **1710** and can be loosely received within the outer cavity **1816**. According to example aspects, the cushioning sheet **700** can define an inner cavity **1826**, which can include portions of the outer cavity **1816**. The cushioned mailer **1700** can be configurable in a folded configuration, as shown in FIGS. **17** and **18**, and an unfolded configuration, as shown in FIG. **19**. The cushioned mailer **1700** can further be configurable in an open orientation, as shown in FIGS. **17** and **18**, and a closed orientation (not shown).

In the folded configuration, the cushioned mailer **1700** can define a front panel **1730** and a rear panel **1740** opposite the front panel **1730**. The cushioning sheet **700** can define a front inner portion **1932** (shown in FIG. **19**) of the front panel **1730**, and the outer sheet **1710** can define front outer portion **1734** of the front panel **1730** opposite the front inner portion **1932**. Similarly, the cushioning sheet **700** (i.e., the inner sheet **1920**) can define a rear inner portion **1942** (shown in FIG. **19**) of the rear panel **1740**, and the outer sheet **1710** can define a rear outer portion **1744** of the rear panel **1740** opposite the rear inner portion **1942**. The front inner portion **1932** of the cushioning sheet **700** can face the rear inner portion **1942** of the cushioning sheet **700**, and the inner cavity **1826** can be substantially defined therebetween.

The inner cavity **1826** can be configured to receive contents **1800** (shown in FIG. **18**) therein, which can be substantially surrounded and cushioned by the cushioning sheet **700**.

Example aspects of the cushioned mailer **1700** can further define a left side **1750**, a right side **1752** opposite the left side **1750**, a bottom end **1754**, and a top end **1756** opposite the bottom end **1754**. The front panel **1730** of the cushioned mailer **1700** can be hingedly coupled to the rear panel **1740** of the cushioned mailer **1700** at the bottom end **1754** at a fold line **1755**, as shown. As such, both the outer cavity **1816** and the inner cavity **1826** can be closed at the bottom end **1754** of the cushioned mailer **1700**. The fold line **1755** can be defined in at least the outer sheet **1710**, and in some aspects, can further be defined in the cushioning sheet **700**. In other aspects, the cushioning sheet **700** can define a cut **1925** (shown in FIG. **19**) at the bottom end **1754** of the cushioned mailer **1700**. Example aspects of the cushioned mailer **1700** can also be sealed at each of the left side **1750** and the right side **1752**. For example, in some aspects, the inner surface **1712** of the outer sheet **1710** can be sealed to itself by an adhesive at the left and right sides **1750**,**1752**, thereby closing the outer cavity **1816** and the inner cavity **1826** at the left and right sides **1750**,**1752**. The adhesive can be, for example, glue or any other suitable adhesive known in the art. In other aspects, the left and right sides **1750**,**1752** can be sealed by any other suitable fastener. Furthermore, in other aspects, the cushioning sheet **700** can be sealed to itself or to the outer sheet **1710** to close the outer cavity **1816** and/or inner cavity **1826** at the left and right sides **1750**, **1752**.

Example aspects of the cushioned mailer **1700** can further be oriented in the open orientation, as shown, and the closed orientation. In the open orientation, an opening **1760** can be defined at the top end **1756** of the cushioned mailer **1700** to allow access to inner cavity **1826**. In the closed orientation, the inner cavity **1826** can be selectively closed at the top end **1756**. In some aspects, the cushioned mailer **1700** can define a closure flap **1770** for selectively covering the opening **1760** and sealing the top end **1756** of the cushioned mailer **1700** in the closed orientation, thereby securing the contents **1800** within the inner cavity **1826**. In the present aspect, the rear outer portion **1744** of the rear panel **1740** of the cushioned mailer **1700** can define the closure flap **1770** extending from the top end **1756** thereof. Thus, the closure flap **1770** can be defined by the outer sheet **1710**, as shown. The closure flap **1770** can define an adhesive strip **1772** extending substantially from the left side **1750** to the right side **1752** of the cushioned mailer **1700**. The adhesive strip **1772** can be selectively covered by a peelable backing **1774** in some aspects, which can be peeled away from the cushioned mailer **1700** to reveal the adhesive strip **1772**. With the peelable backing **1774** removed, the closure flap **1770** can be folded over the top end **1756** of the cushioned mailer **1700**, and the adhesive strip **1772** can be adhered to the front outer portion **1734** of the front panel **1730** of the cushioned mailer **1700**. FIG. **18** illustrates the contents **1800** (for example, a small box **1805**, as shown) being inserted into the inner cavity **1826** of the cushioned mailer **1700** through the top end **1756** thereof in the open orientation.

FIG. **19** illustrates the cushioned mailer **1700** in the unfolded configuration. As shown, the cushioning sheet **700** can be adhered to the inner surface **1712** of the outer sheet **1710**. In the present aspect, the first layer **100** (shown in FIG. **14**) of the cushioning sheet **700** can be adhered to the outer sheet **1710**, and the second layer **200** of the cushioning sheet **700** can face away from the outer sheet **1710**. As such, the second layer **200** of the cushioning sheet **700** can at least partially surround and define the inner cavity **1826** (shown in FIG. **18**) in the folded configuration. The cushioning sheet **700** can further comprise the cushioning inserts **400** received in the voids **830** (shown in FIG. **8**) between the first layer **100** and the second layer **200**, and the cushioning inserts **400** can provide cushioned protection to the contents **1800** (shown in FIG. **18**) received in the inner cavity **1826**. The cushioning sheet **700** can further define the cut **1925** at the bottom end **1754** of the cushioned mailer **1700** to separate the cushioning sheet **700** into the front inner portion **1932** and the rear inner portion **1942**. In other aspects, the cushioning sheet **700** can define the fold line **1755** (shown in FIG. **17**) between the front inner portion **1932** and the rear inner portion **1942**. The cut **1925** can facilitate folding the cushioned mailer **1700** from the unfolded configuration to the folded configuration. As shown, in example aspects, the outer sheet **1710** can define left and right sealing flaps **1930**a,b extending beyond the cushioning sheet **700** at the left and right sides **1750**,**1752** of the cushioned mailer **1700**. In the folded orientation, the portions of the left and right sealing flaps **1930***a,b* proximate to the front inner portion **1932** can be coupled to the portions of the left and right sealing flaps **1930***a,b,* respectively, proximate to the rear inner portion **1942** to seal the left and right sides **1750**,**1752** of the cushioned mailer **1700**.

One should note that conditional language, such as, among others, "can," "could," "might," or "may," unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more particular embodiments or that one or more particular embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment.

It should be emphasized that the above-described embodiments are merely possible examples of implementations, merely set forth for a clear understanding of the principles of the present disclosure. Any process descriptions or blocks in flow diagrams should be understood as representing modules, segments, or portions of code which include one or more executable instructions for implementing specific logical functions or steps in the process, and alternate implementations are included in which functions may not be included or executed at all, may be executed out of order from that shown or discussed, including substantially concurrently or in reverse order, depending on the functionality involved, as would be understood by those reasonably skilled in the art of the present disclosure. Many variations and modifications may be made to the above-described embodiment(s) without departing substantially from the spirit and principles of the present disclosure. Further, the scope of the present disclosure is intended to cover any and all combinations and sub-combinations of all elements, features, and aspects discussed above. All such modifications and variations are intended to be included herein within the scope of the present disclosure, and all possible claims to individual aspects or combinations of elements or steps are intended to be supported by the present disclosure.

That which is claimed is:

1. A cushioning sheet comprising:

a first layer comprising a first base and a plurality of first sockets extending from the first base, each of the first sockets spaced apart from adjacent ones of the first sockets;

a second layer coupled to the first layer, the second layer comprising a second base and a plurality of second sockets extending from the second base, the second base coupled to the first base by a first adhesive to form the cushioning sheet, each of the second sockets aligned with a corresponding one of the first sockets to define a void therebetween; and

a plurality of cushioning inserts, each of the plurality of cushioning inserts substantially spherical in shape and received in a corresponding one of the voids, wherein each void has a diameter that is larger than a diameter of the cushioning insert disposed therein so that each void is configured to allow unrestricted movement of the cushioning insert disposed therein about the corresponding void;

wherein:

each of the first sockets are arranged in a plurality of linear rows extending in a lateral direction and linear columns extending in a longitudinal direction perpendicular to the lateral direction;

in each of the linear rows, none of the first sockets are longitudinally aligned with any of the first sockets in adjacent linear rows;

in each of the linear columns, none of the first sockets are laterally aligned with any of the first sockets in adjacent linear columns; and

each of the first layer and the second layer comprise a coating configured to limit the passage of air therethrough.

2. The cushioning sheet of claim **1**, wherein each of the plurality of cushioning inserts comprises a starch material, and wherein the starch material is repulpable.

3. The cushioning sheet of claim **1**, wherein at least the first adhesive is a starch paste, and wherein the starch paste is repulpable.

4. The cushioning sheet of claim **1**, wherein:

each of the first sockets is formed as a substantially semi-spherical dome defining a first recess and a first opening allowing access to the first recess;

each of the second sockets is formed as a substantially semi-spherical dome defining a second recess and a second opening allowing access to the second recess; and

each of the cushioning inserts is at least partially received in one of the first recesses and at least partially received in one of the second recesses.

5. The cushioning sheet of claim **1**, wherein:

the first base is substantially planar and defines a first upper surface and a first lower surface opposite the first upper surface;

the second base is substantially planar and defines a second upper surface and a second lower surface; and

the second upper surface of the second base is coupled to the first lower surface of the first base.

6. The cushioning sheet of claim **1**, wherein each of the first layer and the second layer comprises a paper material.

7. The cushioning sheet of claim **1**, wherein each of the first layer and the second layer comprises stretchy paper material.

8. The cushioning sheet of claim **1**, wherein the cushioning sheet is repulpable.

9. The cushioning sheet of claim **8**, wherein the coating is configured to prevent the passage of air therethrough.

10. A cushioned mailer comprising:

a first layer defining a plurality of first sockets;

a second layer defining a plurality of second sockets, each of the first sockets aligned with a corresponding one of the second sockets to define a void therebetween; and

a cushioning insert received within each of the voids, wherein each void has a diameter that is larger than a diameter of the cushioning insert disposed therein so that each void is configured to allow unrestricted movement of the cushioning insert disposed therein about the corresponding void;

wherein:

the first layer is coupled to the second layer by a first adhesive to define a cushioning sheet;

the cushioning sheet defines an inner cavity configured to receive contents;

the inner cavity is at least partially surrounded by the second layer;

the first layer faces away from the inner cavity;

the cushioned mailer further comprises an outer sheet defining an outer cavity;

the cushioning sheet is received within the outer cavity;

each of the first sockets are arranged in a plurality of linear rows extending in a lateral direction and linear columns extending in a longitudinal direction perpendicular to the lateral direction;

in each of the linear rows, none of the first sockets are longitudinally aligned with any of the first sockets in adjacent linear rows;

in each of the linear columns, none of the first sockets are laterally aligned with any of the first sockets in adjacent linear columns; and

each of the first layer and the second layer comprise a coating configured to limit the passage of air therethrough.

11. The cushioned mailer of claim **10**, wherein:

the cushioned mailer defines a left side, a right side opposite the left side, a bottom end, and a top end opposite the bottom end;

the inner cavity is closed at the bottom end, the left side, and the right side; and

the cushioned mailer defining an opening at the top end allowing access to the inner cavity.

12. The cushioned mailer of claim **11**, further comprises a closure flap extending from the top end of the cushioned mailer and configured to selectively cover the opening and to close the inner cavity at the top end, the closure flap comprising an adhesive strip and a peelable backing selectively covering the adhesive strip.

13. The cushioned mailer of claim **10**, wherein the outer sheet defines an inner surface and an outer surface, the inner surface defines the outer cavity, and the first layer of the cushioned sheet is coupled to the inner surface.

14. The cushioned mailer of claim **10**, wherein:

the cushioning sheet defines a front inner portion of a front panel of the cushioned mailer and a rear inner portion of a rear panel of the cushioned mailer, the front inner portion substantially facing the rear inner portion, the inner cavity defined between the front inner portion and the rear inner portion;

the outer sheet defines a front outer portion of the front panel and a rear outer portion of the rear panel; and

the front panel is folded relative to the rear panel at the fold line at the bottom end of the cushioned mailer.

15. The cushioned mailer of claim **14**, wherein the front outer portion is sealed with the rear outer portion at a left side of the cushioned mailer and a right side of the cushioned mailer, and wherein a top end of the cushioned mailer defines an opening allowing access to the inner cavity.

16. The cushioned mailer of claim **10**, wherein the coating is configured to prevent the passage of air therethrough.

17. The cushioned mailer of claim **10**, wherein each of the first layer and the second layer comprises stretchy paper material.

18. A cushioning sheet comprising:

a first layer comprising a first base and a plurality of first sockets extending from the first base, each of the first sockets spaced apart from adjacent ones of the first sockets;

a second layer coupled to the first layer, the second layer comprising a second base and a plurality of second sockets extending from the second base, the second base coupled to the first base by an adhesive to form the cushioning sheet, each of the second sockets aligned with a corresponding one of the first sockets to define a void therebetween; and

a plurality of cushioning inserts, each of the plurality of cushioning inserts substantially spherical in shape and received in a corresponding one of the voids;

wherein each void has a diameter that is larger than a diameter of the cushioning insert disposed therein so that each void is configured to allow unrestricted movement of the cushioning insert disposed therein about the corresponding void.

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